PTERYGOPALATINE FOSSA AND GANGLION

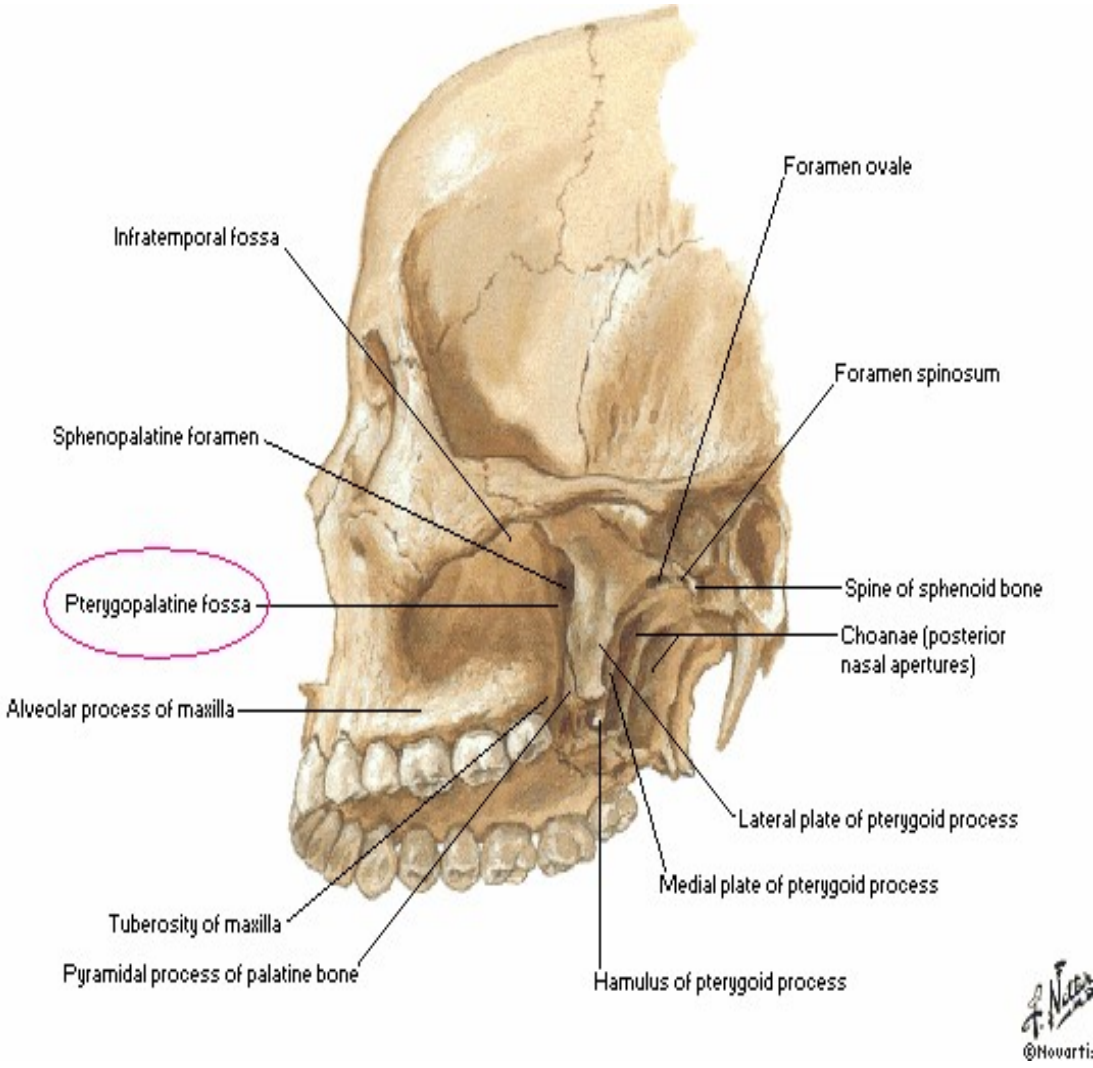
LEARNING OBJECTIVES

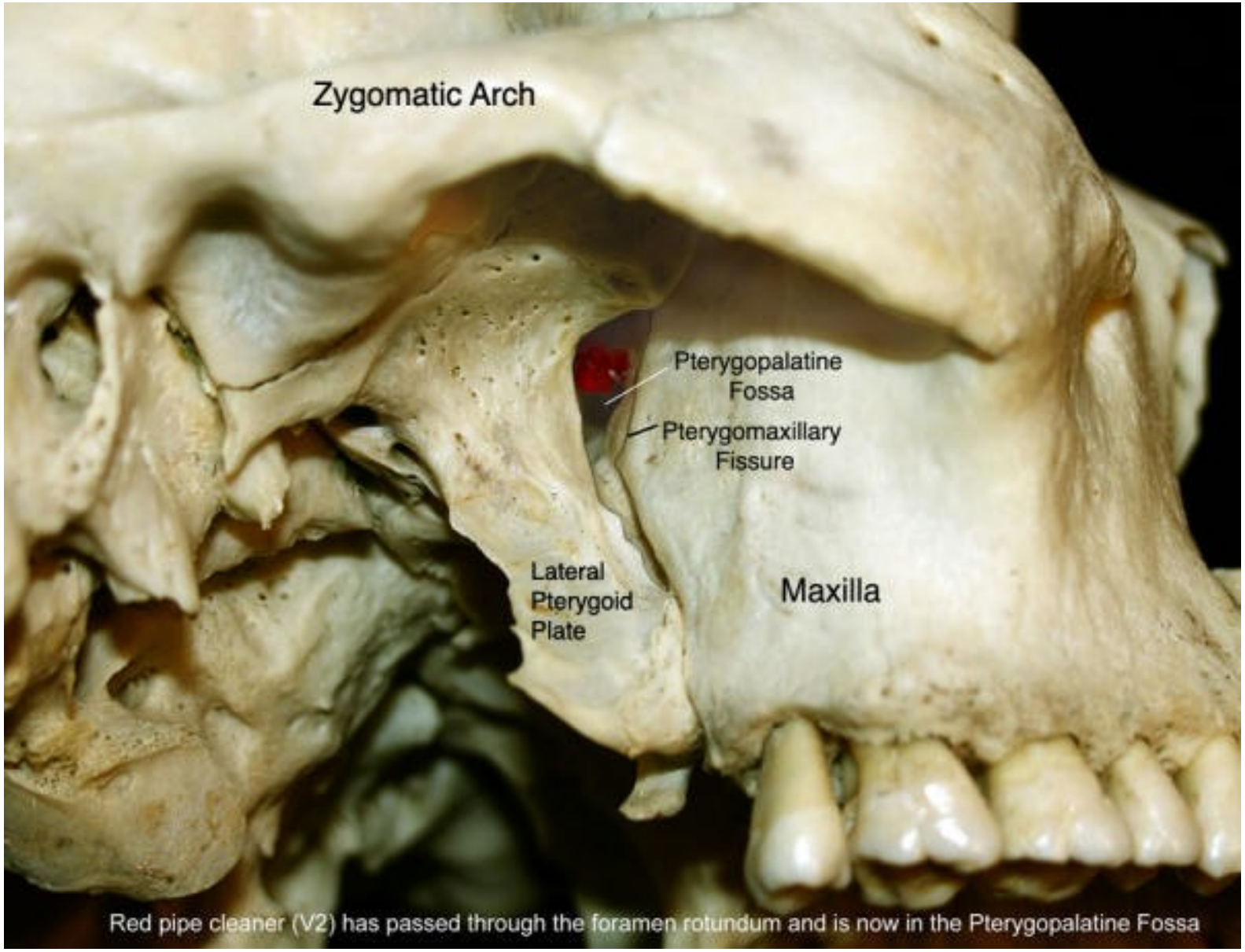
- At the end of the lecture, students should be able to:
- Explain the Boundaries of pterygopalatine fossa.
- Enumerate the Contents of pterygopalatine fossa and their relative description.
- Discuss about the openings in pterygopalatine fossa.
- Discuss the Connections and Branches of pterygopalatine ganglion.

PTERYGOPALATINE FOSSA.

- It is a small space between the back of the maxilla and the front of the pterygoid process of the sphenoid bone.

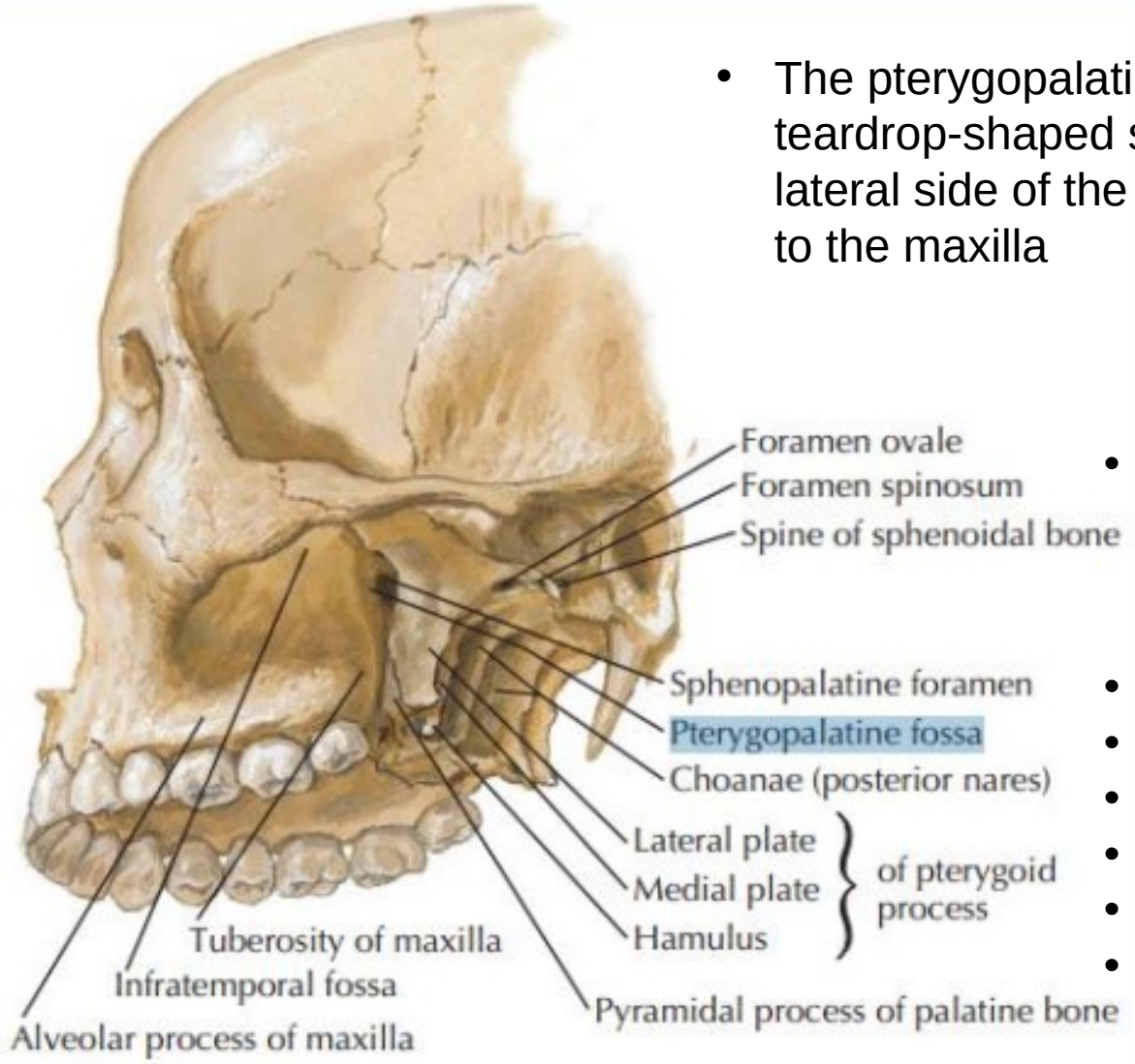
- Concerned with the blood and nerve supply of the upper jaw including nose and lacrimal gland.





Red pipe cleaner (V2) has passed through the foramen rotundum and is now in the Pterygopalatine Fossa

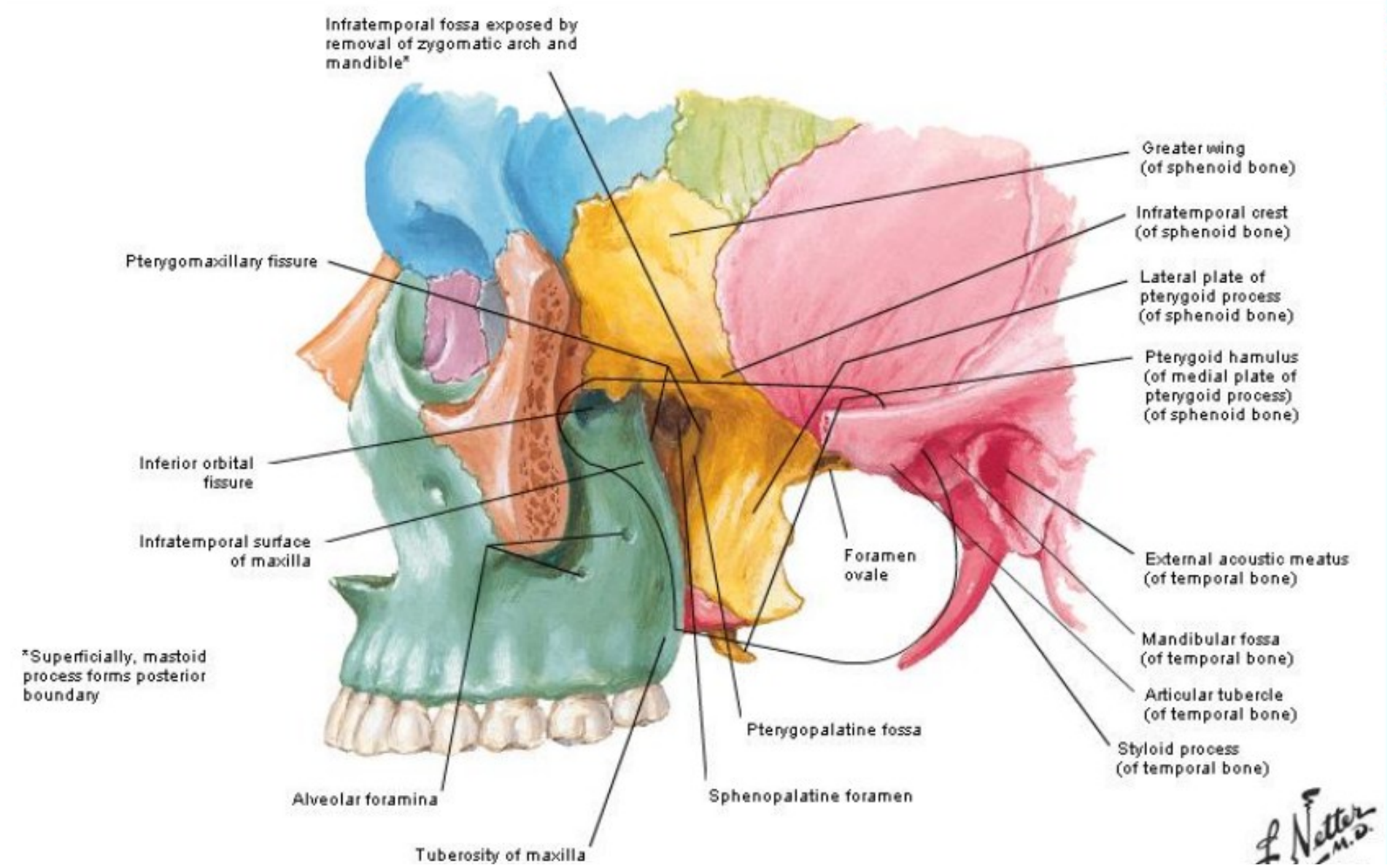
PTERYGOPALATINE FOSSA.



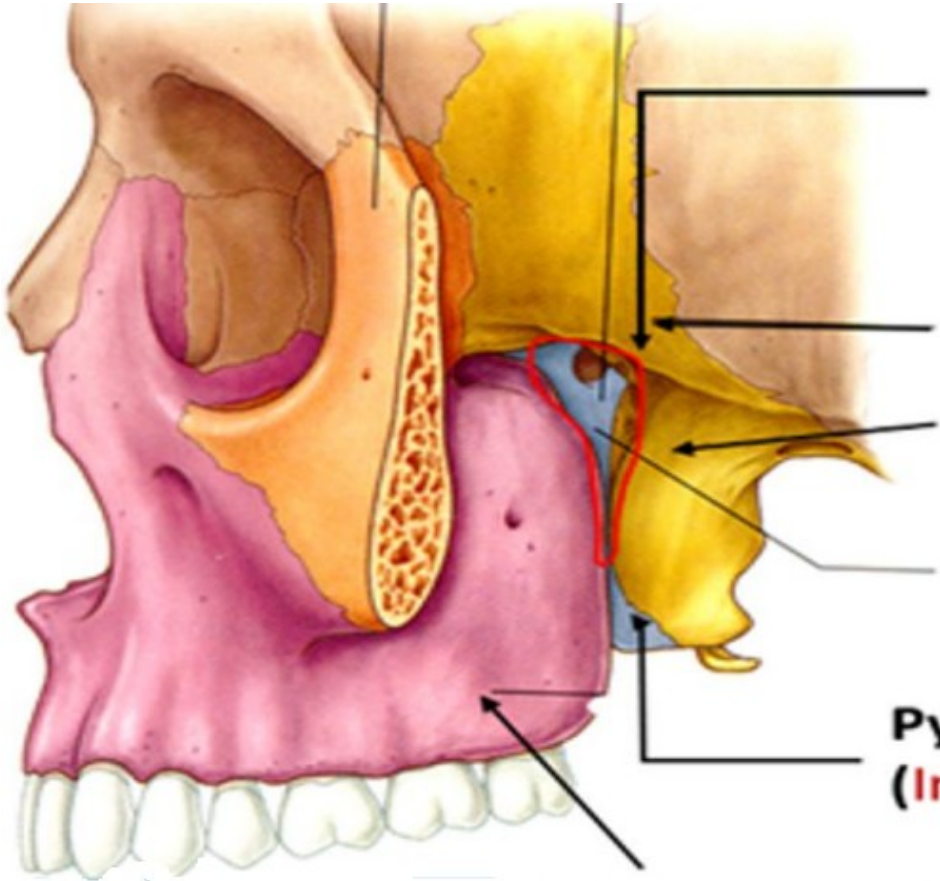
- The pterygopalatine fossa is an inverted teardrop-shaped space between bones on the lateral side of the skull immediately posterior to the maxilla

- Although small in size, the pterygopalatine fossa communicates via fissures and foramina in its walls with the:
- middle cranial fossa;
- infratemporal fossa;
- floor of the orbit;
- lateral wall of the nasal cavity;
- oropharynx; and
- roof of the oral cavity.

BOUNDARIES.


Infratemporal fossa exposed by removal of zygomatic arch and mandible*
Greater wing (of sphenoid bone)
Infratemporal crest (of sphenoid bone)
Pterygomaxillary fissure
Lateral plate of pterygoid process (of sphenoid bone)
Pterygoid hamulus (of medial plate of pterygoid process) (of sphenoid bone)
Inferior orbital fissure
Infratemporal surface of maxilla
Foramen ovale
External acoustic meatus (of temporal bone)
Mandibular fossa (of temporal bone)
Articular tubercle (of temporal bone)
Pterygopalatine fossa
Styloid process (of temporal bone)
Alveolar foramina
Sphenopalatine foramen
Tuberosity of maxilla
*Superficially, mastoid process forms posterior boundary

BOUNDARIES



Pterygomaxillary fissure
(Laterally)

Greater wing of sphenoid
(Superiorly)

Pterygoid process
(Posteriorly)

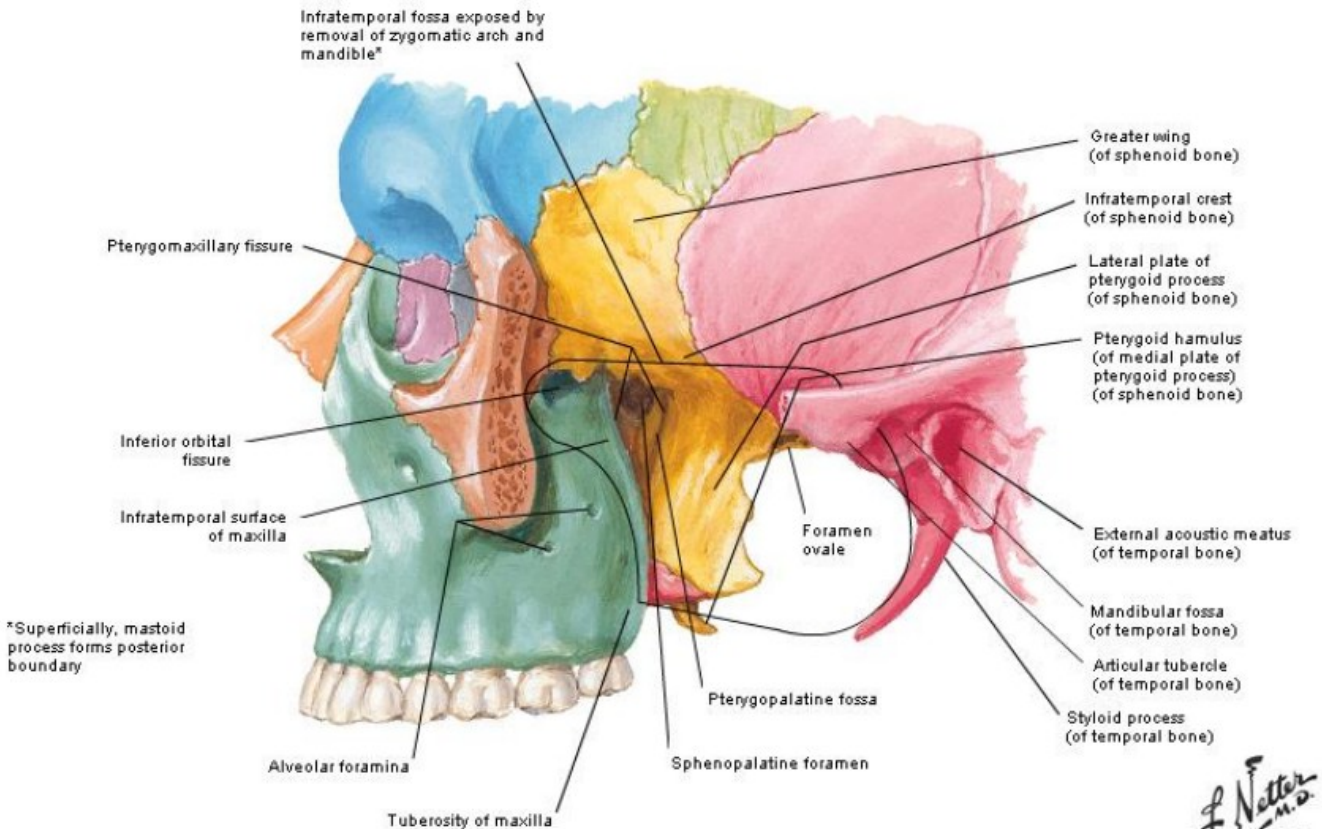
Perpendicular plate of palatine bone (Medially)

Pyramidal process of palatine bone
(Inferiorly)

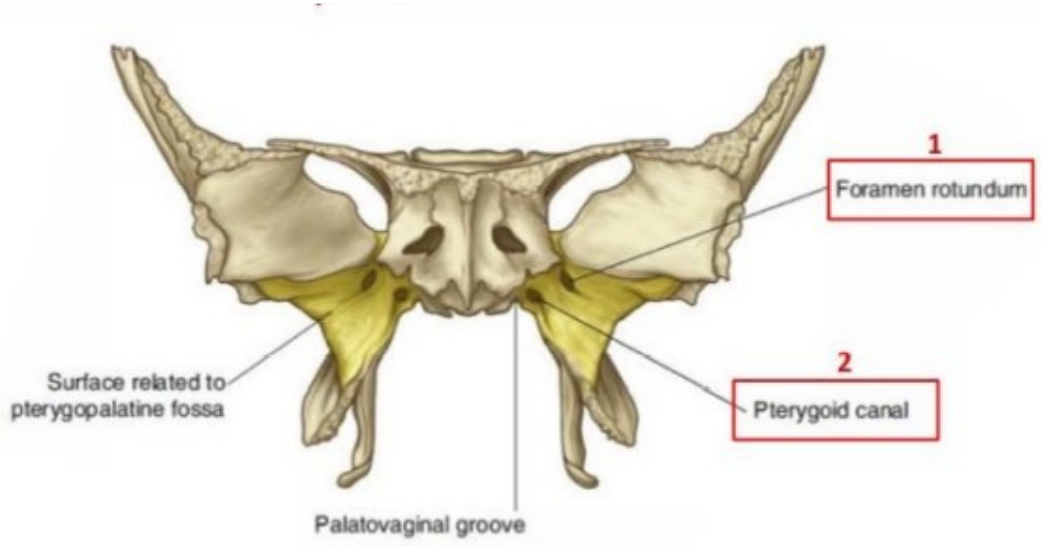
Posterior surface of maxilla
(Anteriorly)

BOUNDARIES.

- **Anterior:** superomedial part of the infratemporal surface of **maxilla.**
- **Laterally:** **Pterygomaxillary fissure** (between maxilla & lateral pterygoid plates).
- **Medially:** **Palatine** bone.
- **Roof:** Body of Sphenoid, Orbital process of palatine bone.
- **Posteriorly:** Sphenoid (root of the pterygoid process)





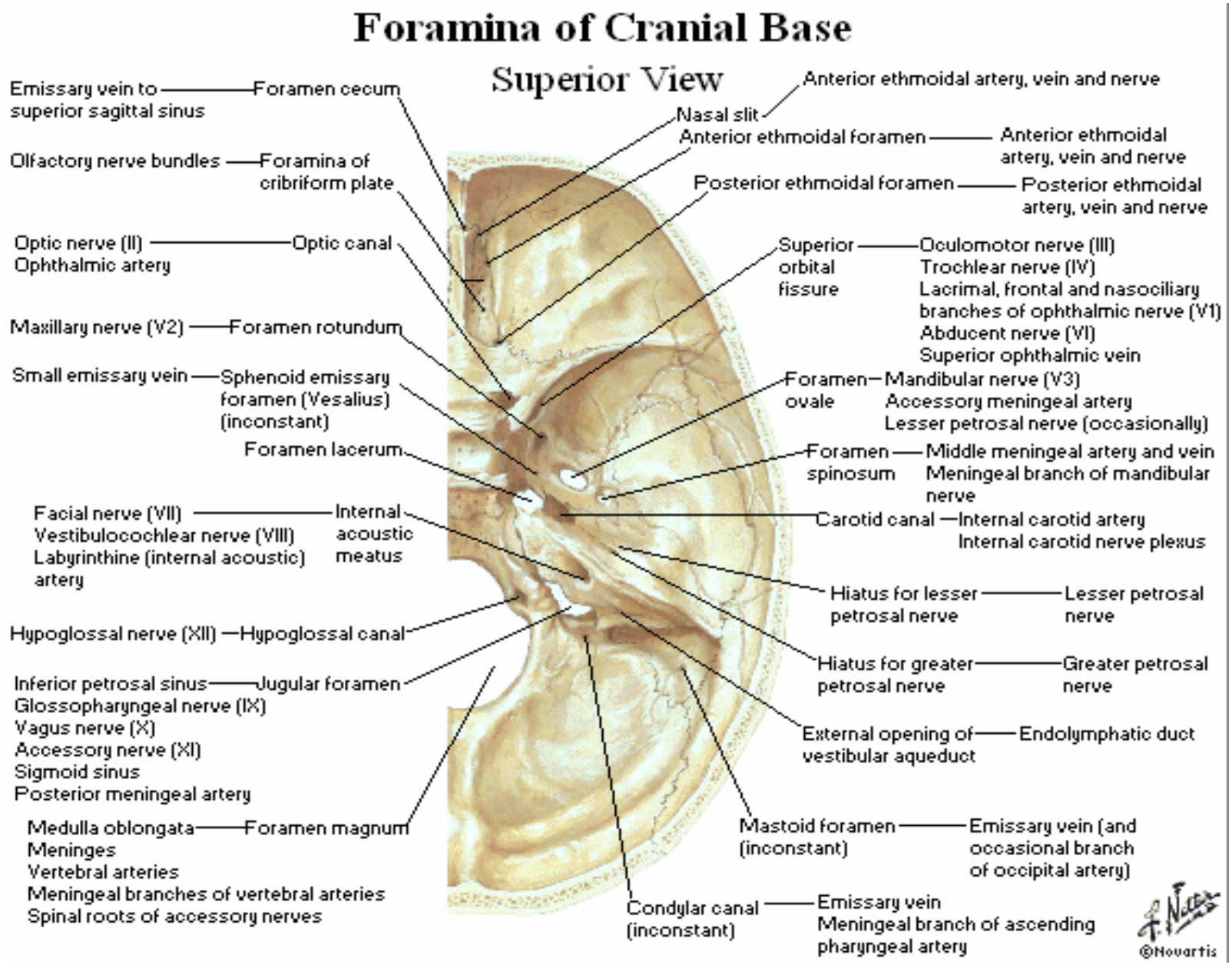


- **Posteriorly** : Sphenoid (root of the pterygoid process)

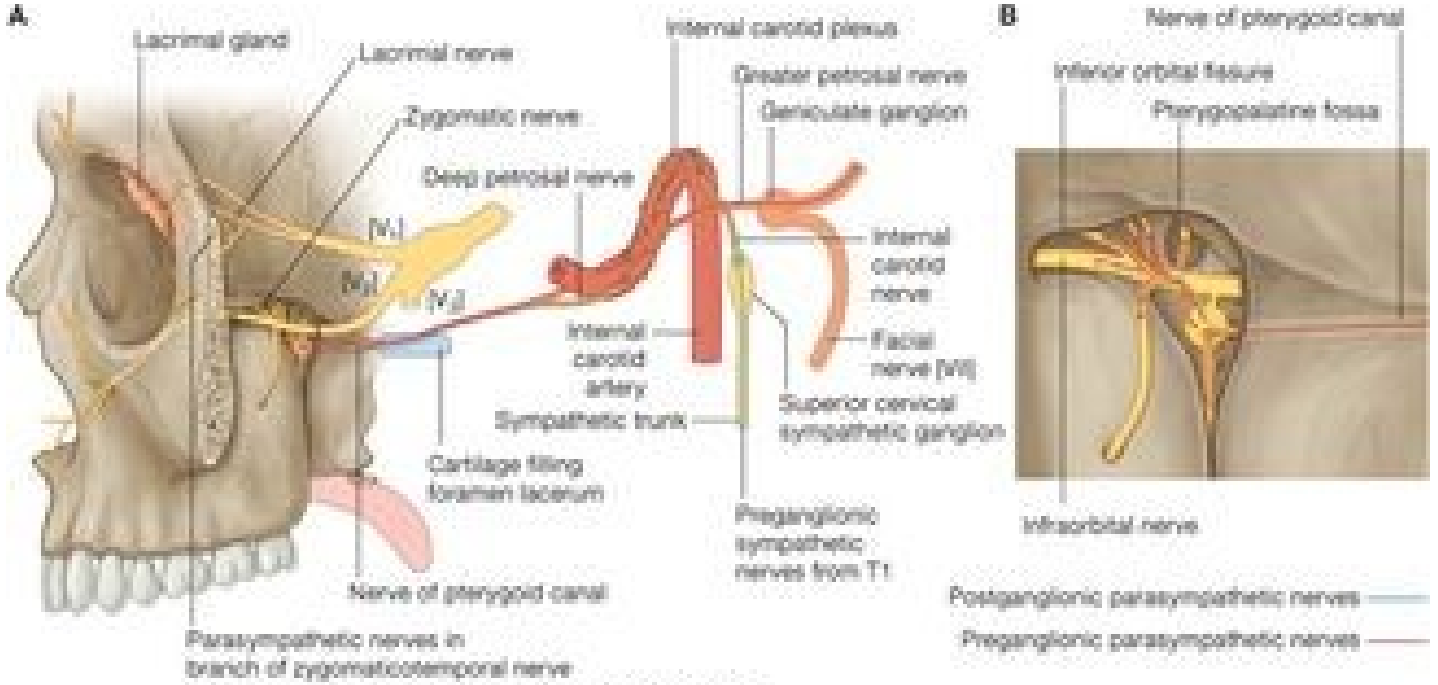
CONNECTIONS

Direction	Passage	Connection
Posteriorly	foramen rotundum	middle cranial fossa
Posteriorly	pterygoid canal (Vidian)	middle cranial fossa, foramen lacerum
Posteriorly	palatovaginal canal (pharyngeal)	nasal cavity/nasopharynx
Anteriorly	inferior orbital fissure	orbit
Medially	sphenopalatine foramen	nasal cavity
Laterally	pterygomaxillary fissure	infratemporal fossa
Inferiorly	greater palatine canal (pterygopalatine)	oral cavity, lesser palatine canals

OPENINGS IN PTERYGOPALATINE FOSSA.



CONTENTS OF PTERYGOPALATINE FOSSA



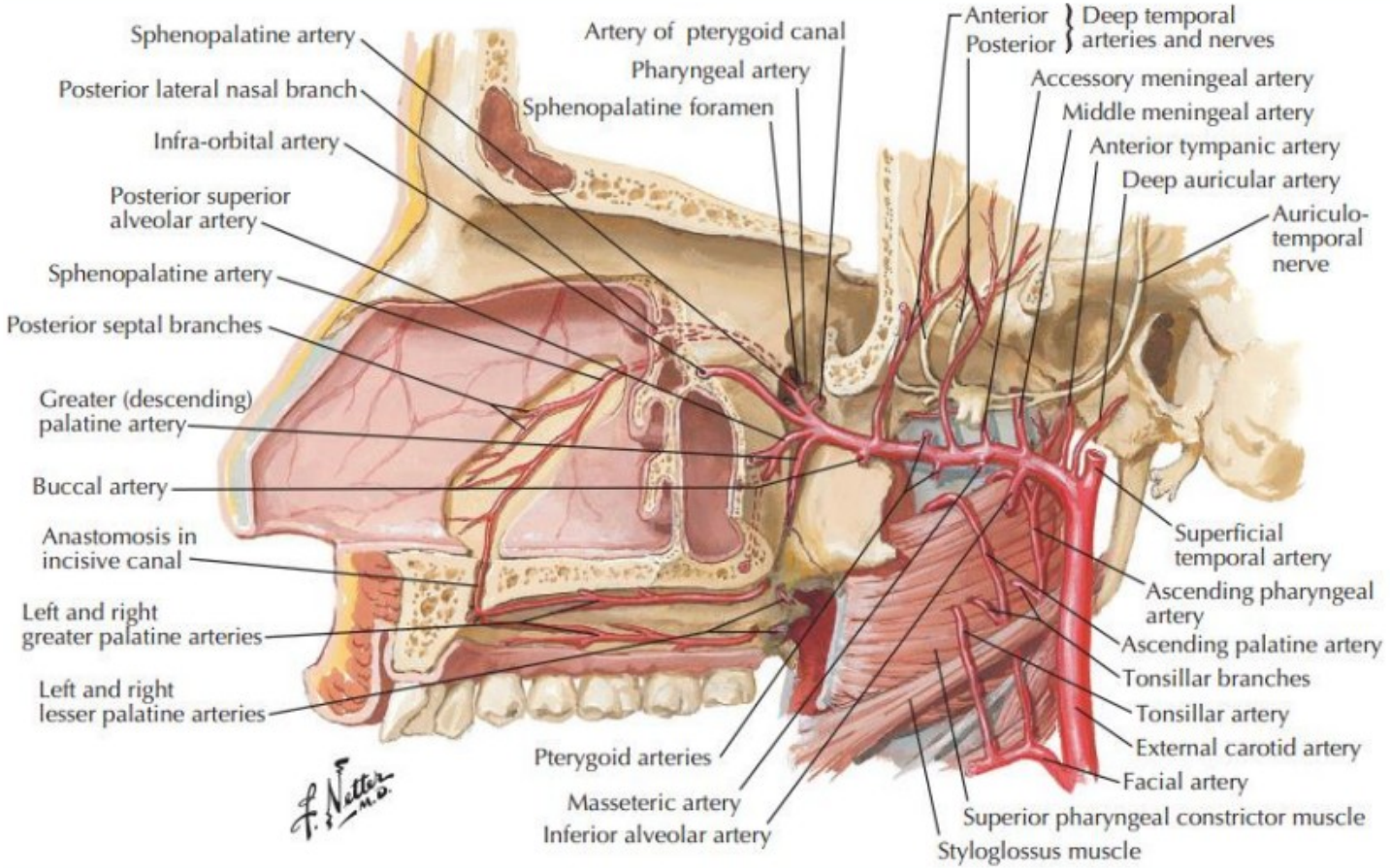
1. Third part of maxillary artery.
2. Maxillary nerve.
3. Pterygopalatine ganglion.
4. In addition, the nerve of the pterygoid canal enters the fossa carrying:

a. Preganglionic parasympathetic fibers from the greater petrosal branch of the facial nerve [VII]; and

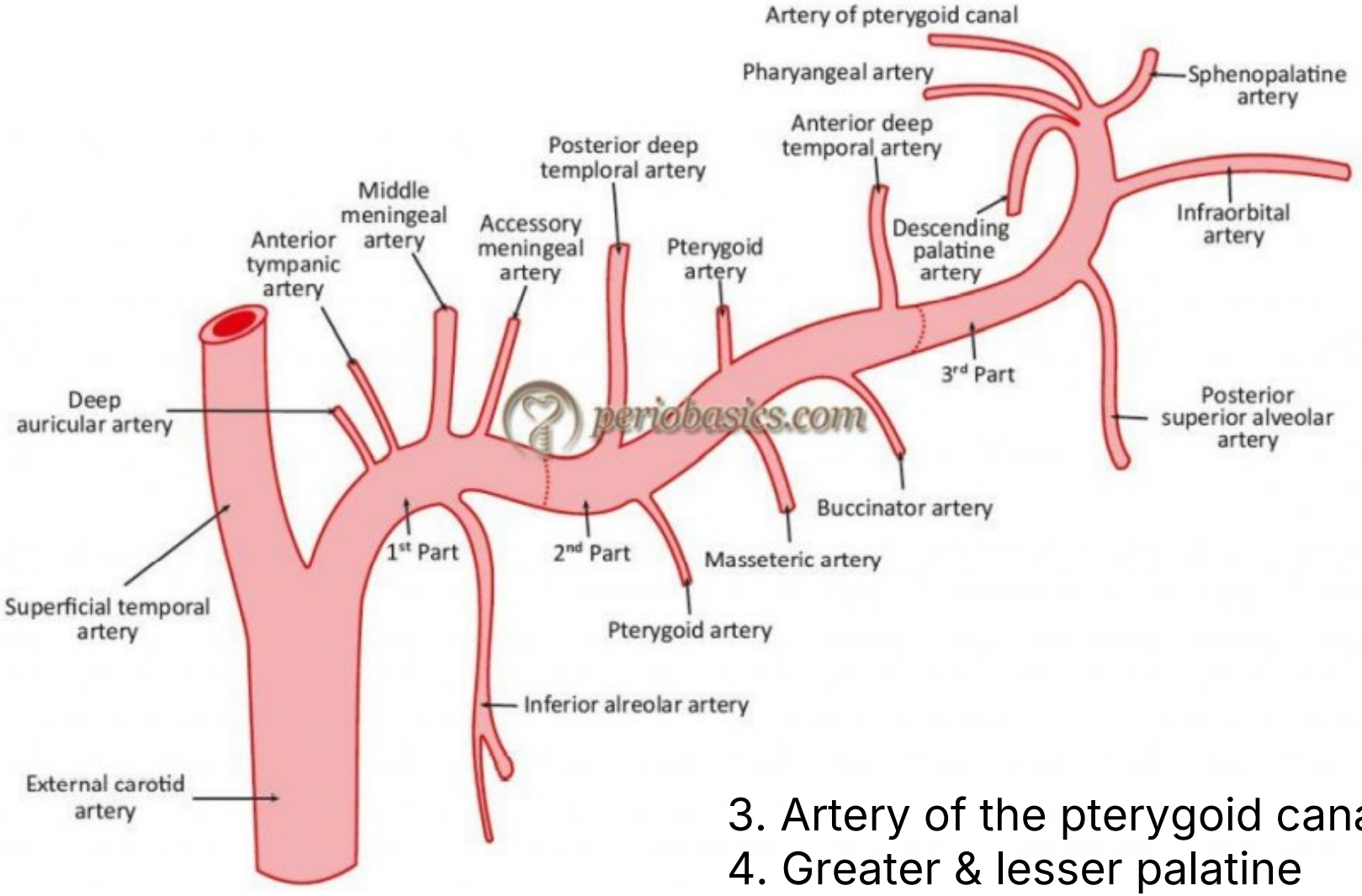
b. Postganglionic sympathetic fibers from the deep petrosal branch of the carotid plexus.

In addition to nerves and arteries, veins and lymphatics also pass through the pterygopalatine fossa

1. THIRD PART OF MAXILLARY ARTERY



- Beyond the lateral pterygoid muscle enters into the pterygopalatine fossa.
- The artery leaves the fossa through the sphenopalatine foramen, changing its name to the sphenopalatine artery and becoming the main artery of the nasal cavity.

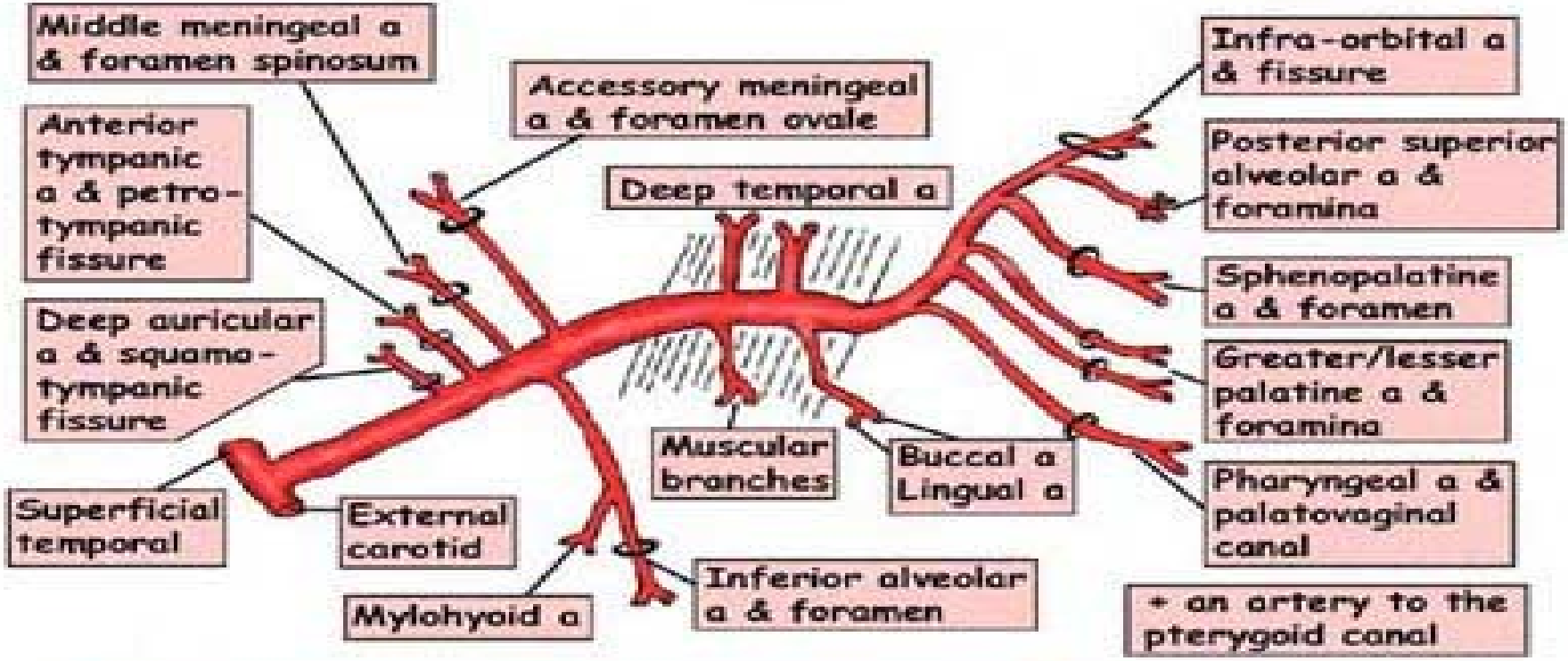


- **Branches:**

1. Infraorbital artery.
2. Posterior superior alveolar artery.
3. Artery of the pterygoid canal.
4. Greater & lesser palatine arteries (Descending palatine).
5. Sphenopalatine artery.
6. Pharyngeal branch.

MAXILLARY ARTERY

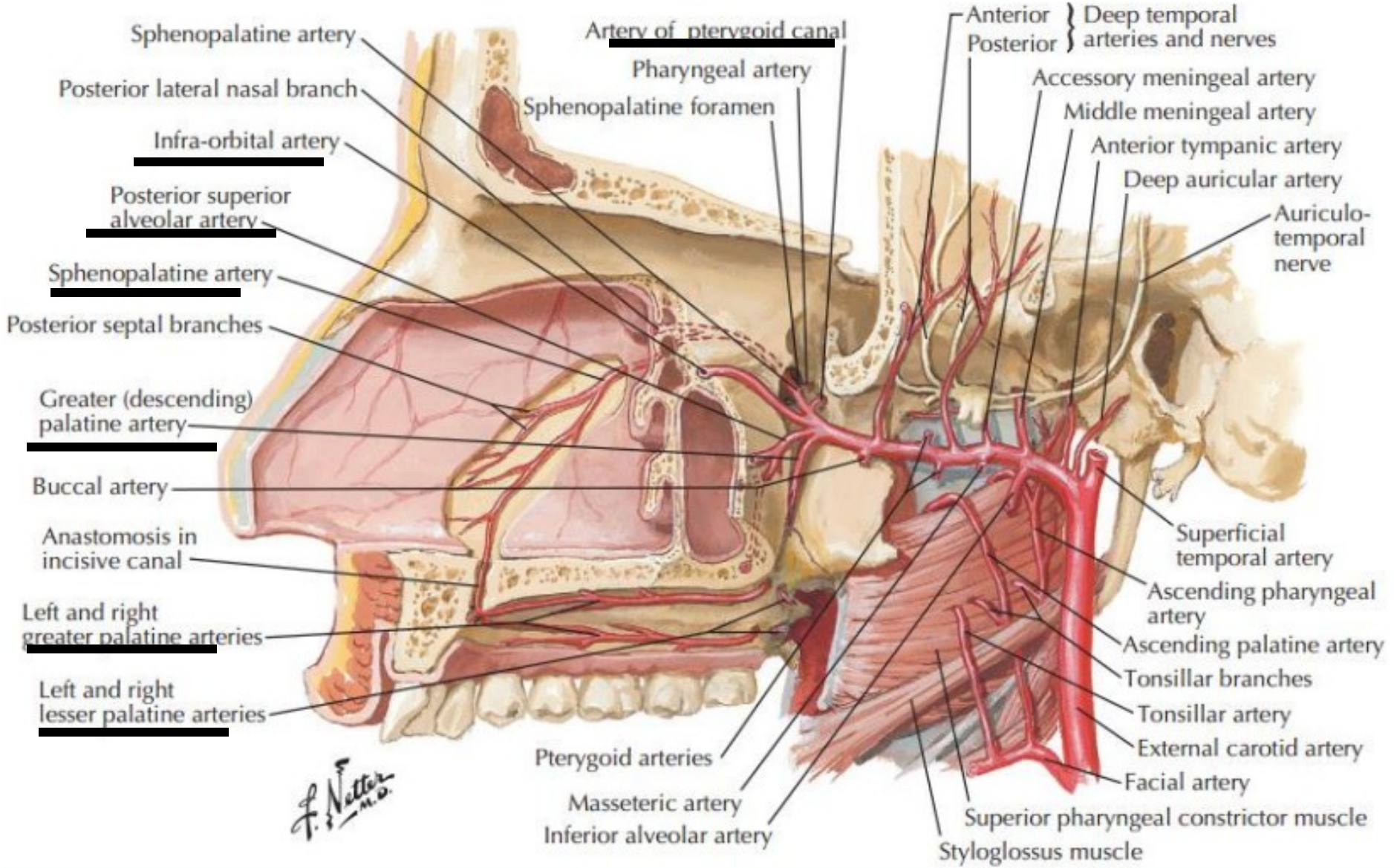
In infratemporal fossa, either within or lateral to the superficial head of lateral pterygoid muscle. This muscle is shown below

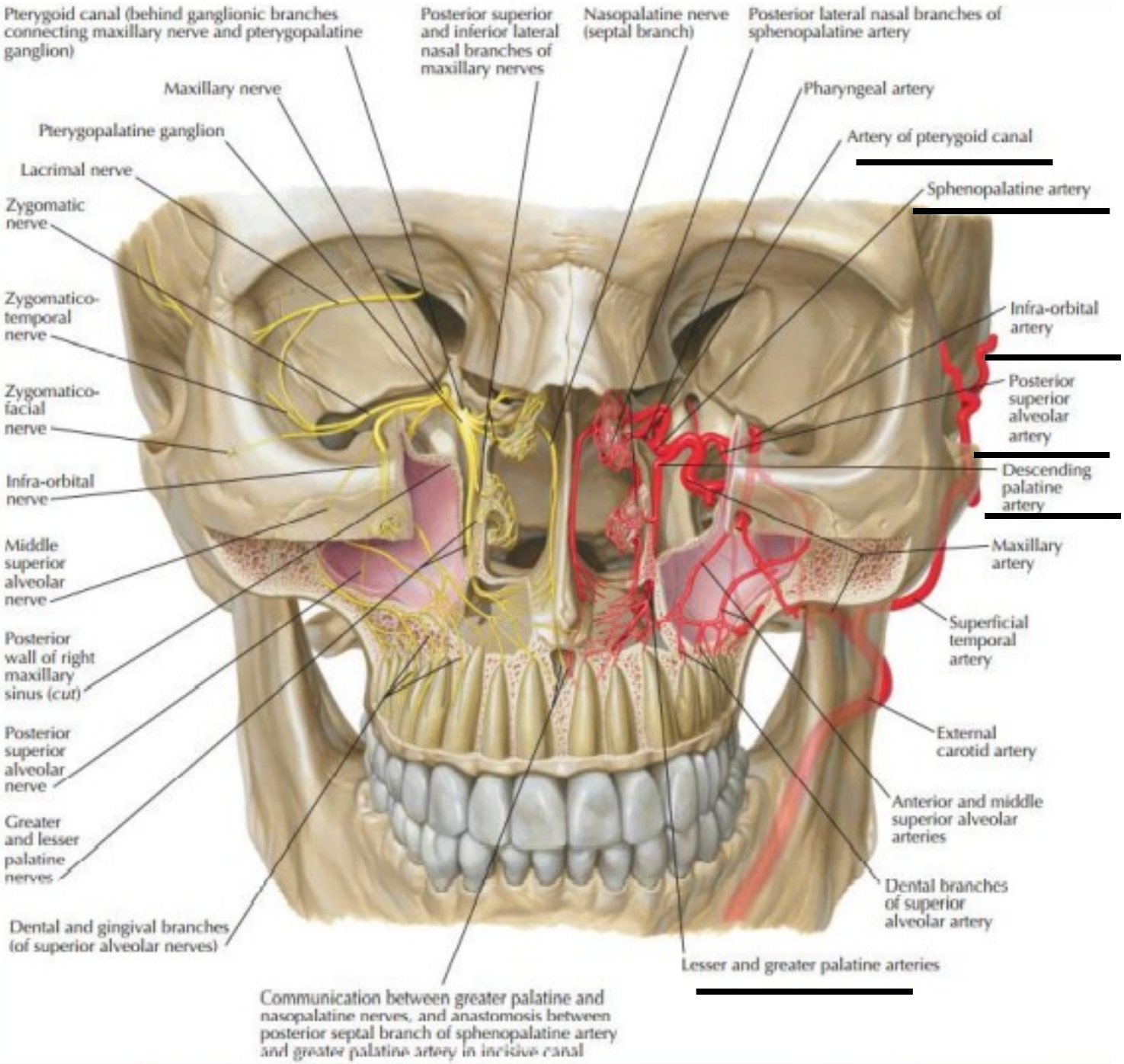


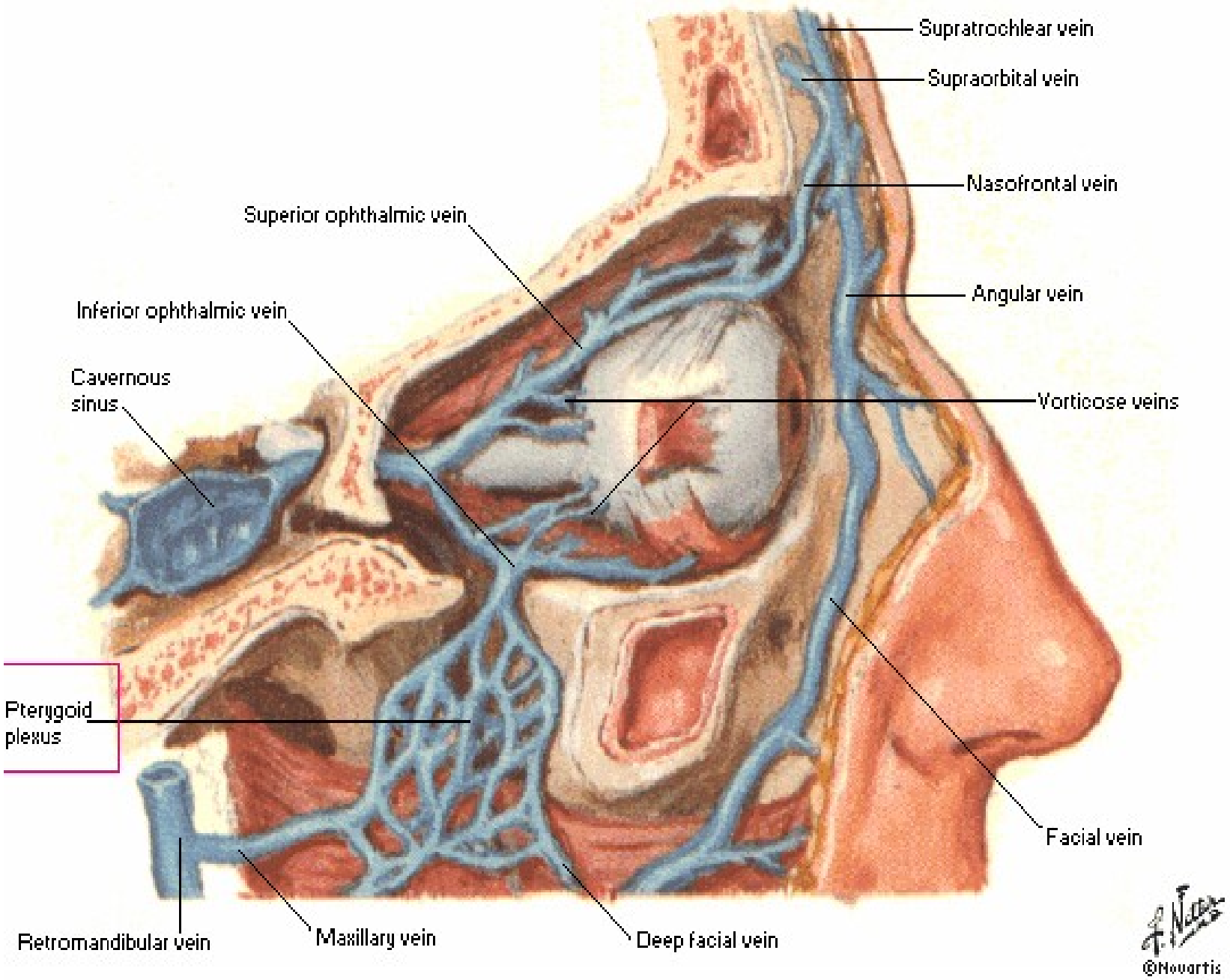
BEFORE LATERAL PTERYGOID 5 BRANCHES INTO BONE

LATERAL OR WITHIN LATERAL PTERYGOID. 4/5 BRANCHES TO SOFT TISSUE

BEYOND LATERAL PTERYGOID 5/6 BRANCHES WITH NERVES

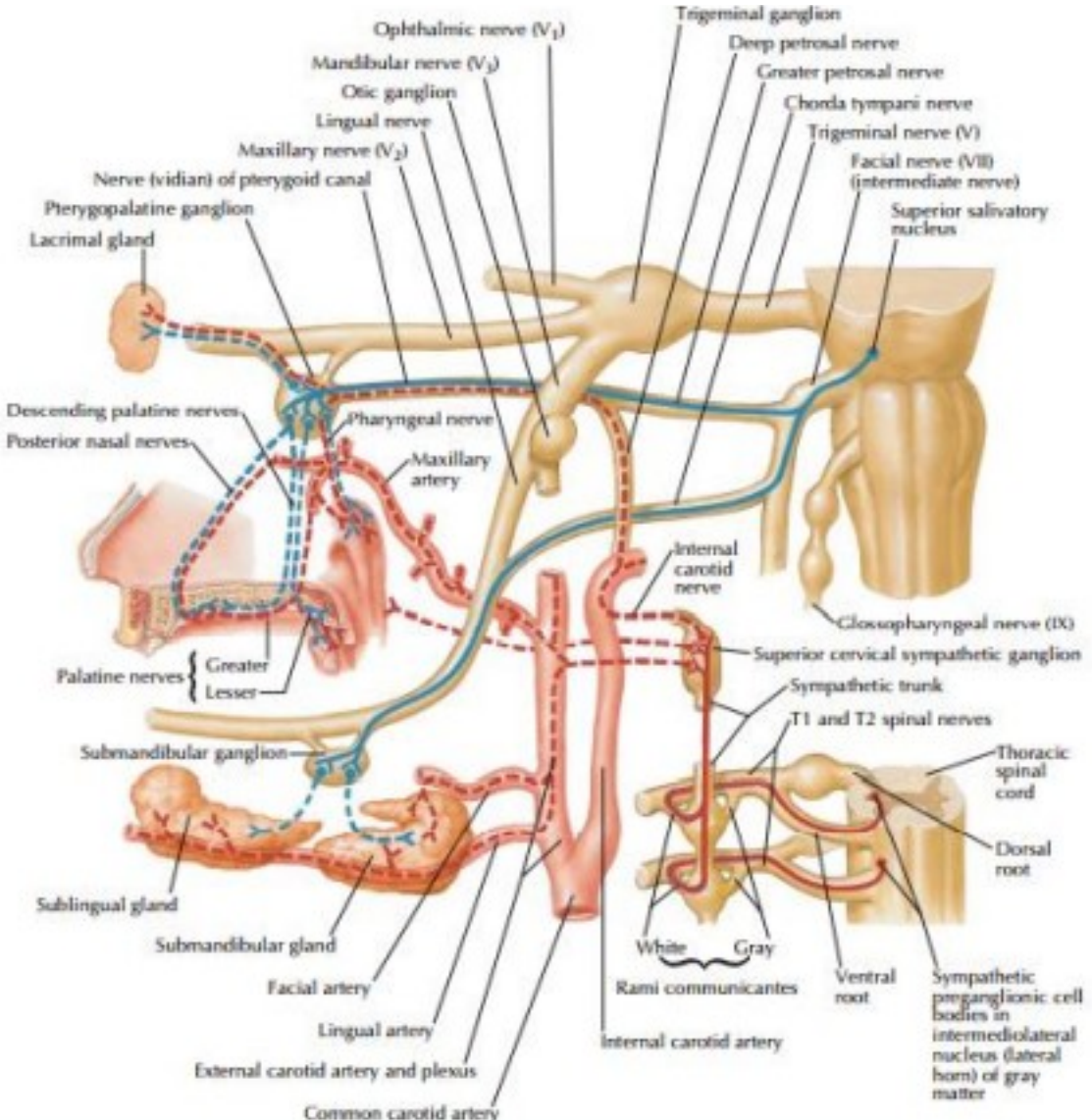

Sphenopalatine artery
Posterior lateral nasal branch
Infra-orbital artery
Posterior superior alveolar artery
Sphenopalatine artery
Posterior septal branches
Greater (descending) palatine artery
Buccal artery
Anastomosis in incisive canal
Left and right greater palatine arteries
Left and right lesser palatine arteries
Artery of pterygoid canal
Pharyngeal artery
Sphenopalatine foramen
Anterior
Posterior
Deep temporal arteries and nerves
Accessory meningeal artery
Middle meningeal artery
Anterior tympanic artery
Deep auricular artery
Auriculo-temporal nerve
Superficial temporal artery
Ascending pharyngeal artery
Ascending palatine artery
Tonsillar branches
Tonsillar artery
External carotid artery
Facial artery
Superior pharyngeal constrictor muscle
Styloglossus muscle
Pterygoid arteries
Masseteric artery
Inferior alveolar artery
F. Netter M.D.

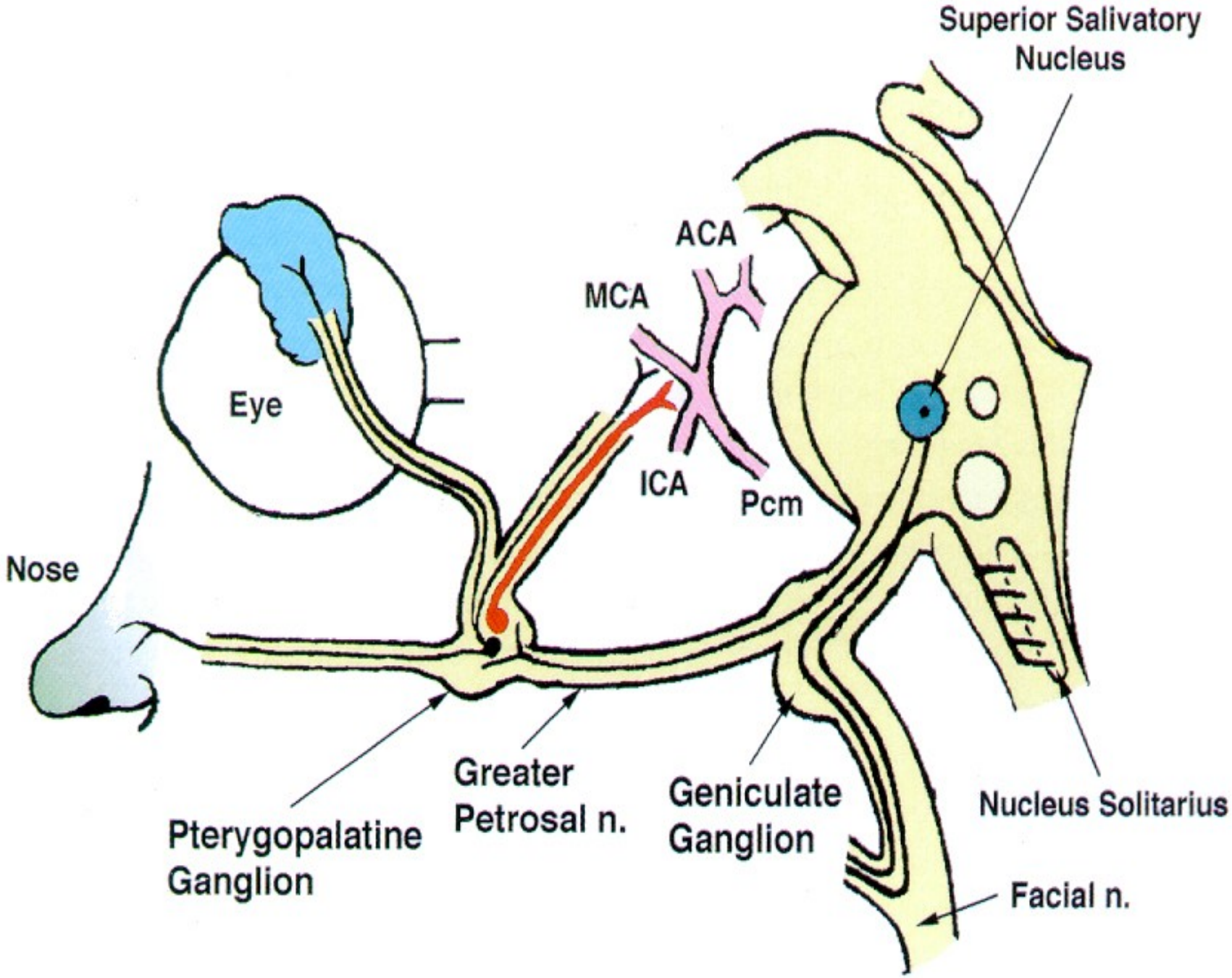

Pterygoid canal (behind ganglionic branches connecting maxillary nerve and pterygopalatine ganglion)
Posterior superior and inferior lateral nasal branches of maxillary nerves
Nasopalatine nerve (septal branch)
Posterior lateral nasal branches of sphenopalatine artery
Maxillary nerve
Pharyngeal artery
Pterygopalatine ganglion
Artery of pterygoid canal
Lacrimal nerve
Sphenopalatine artery
Zygomatic nerve
Zygomatico-temporal nerve
Infra-orbital artery
Posterior superior alveolar artery
Zygomatico-facial nerve
Descending palatine artery
Infra-orbital nerve
Maxillary artery
Middle superior alveolar nerve
Superficial temporal artery
Posterior wall of right maxillary sinus (cut)
External carotid artery
Posterior superior alveolar nerve
Anterior and middle superior alveolar arteries
Greater and lesser palatine nerves
Dental branches of superior alveolar artery
Dental and gingival branches (of superior alveolar nerves)
Lesser and greater palatine arteries
Communication between greater palatine and nasopalatine nerves, and anastomosis between posterior septal branch of sphenopalatine artery and greater palatine artery in incisive canal

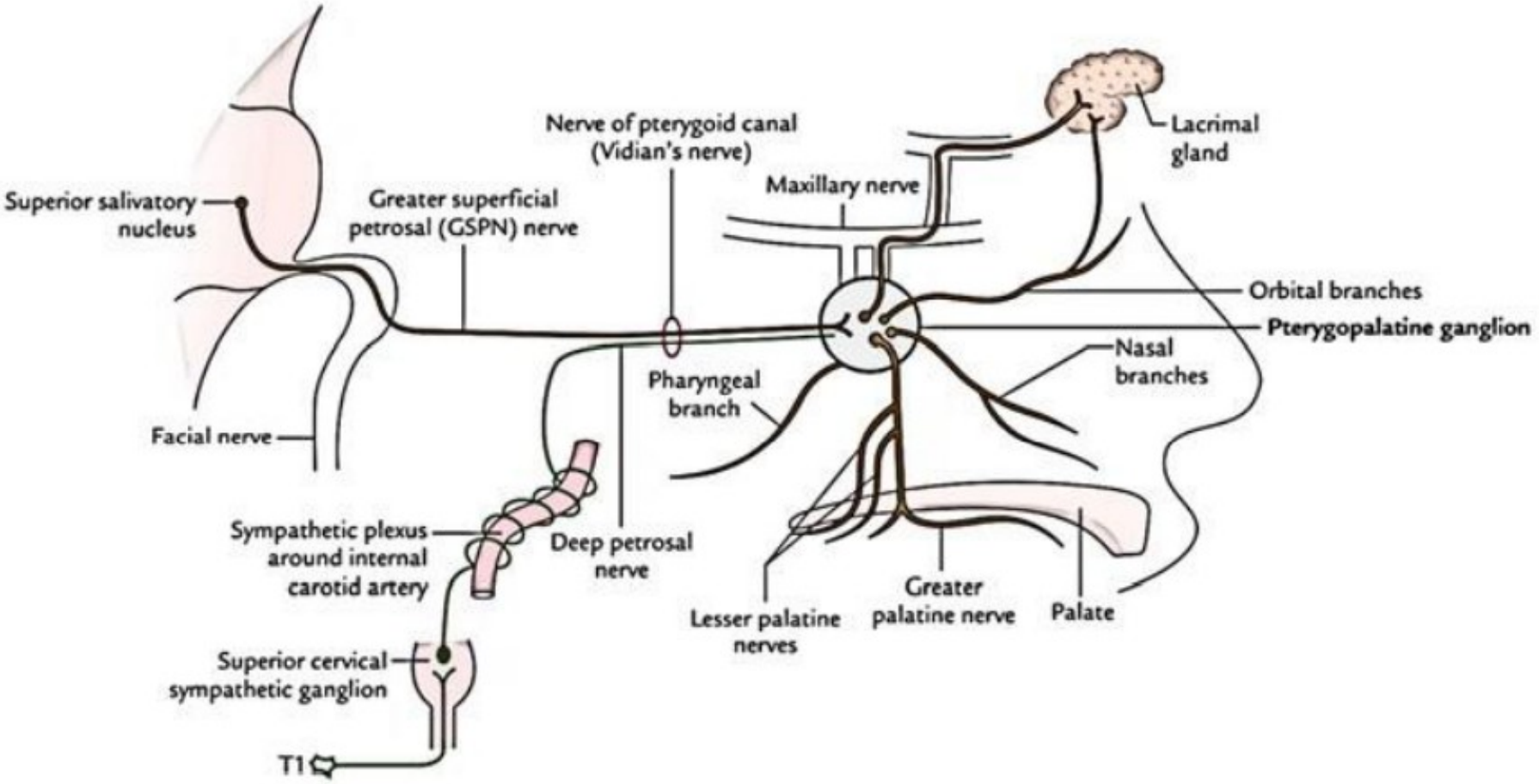

Supratrochlear vein
Supraorbital vein
Nasofrontal vein
Superior ophthalmic vein
Angular vein
Inferior ophthalmic vein
Cavernous sinus
Vorticose veins
Pterygoid plexus
Facial vein
Retromandibular vein
Maxillary vein
Deep facial vein
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2. PTERYGOPALATINE GANGLION.

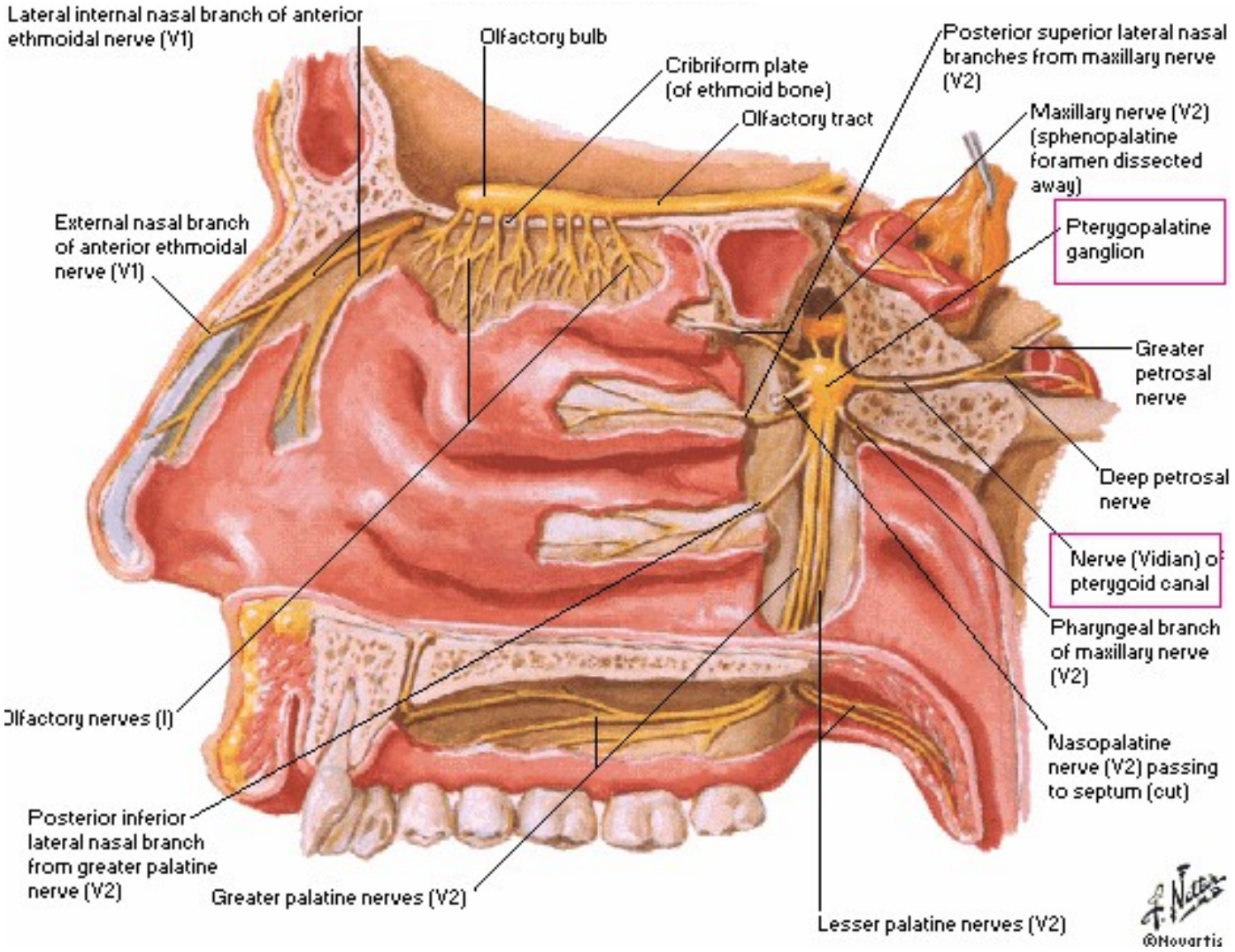
- It is a relay station between the **superior salivary nucleus** in the pons and the **lacrimal gland** and the **glands of palate, nose** and **paranasal sinuses.**
- Also known as the ganglion of **Hay Fever** (running nose and eyes).




Superior Salivatory Nucleus
ACA
MCA
Eye
ICA
Pcm
Nose
Pterygopalatine Ganglion
Greater Petrosal n.
Geniculate Ganglion
Nucleus Solitarius
Facial n.


Nerve of pterygoid canal
(Vidian's nerve)
Lacrimal
gland
Maxillary nerve
Superior salivatory
nucleus
Greater superficial
petrosal (GSPN) nerve
Orbital branches
Pterygopalatine ganglion
Nasal
branches
Pharyngeal
branch
Facial nerve
Sympathetic plexus
around internal
carotid artery
Deep petrosal
nerve
Greater
palatine nerve
Palate
Lesser palatine
nerves
Superior cervical
sympathetic ganglion
T1

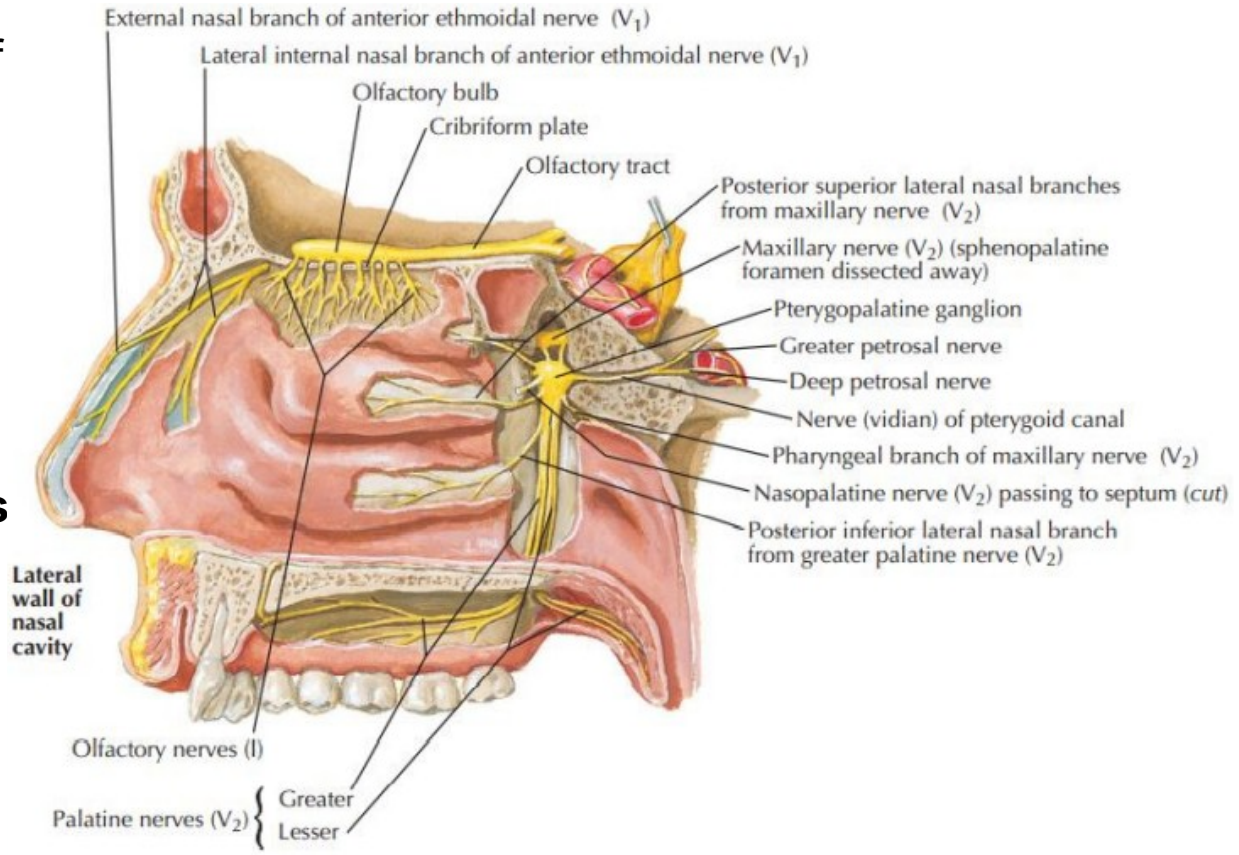
Pterygopalatine ganglion.

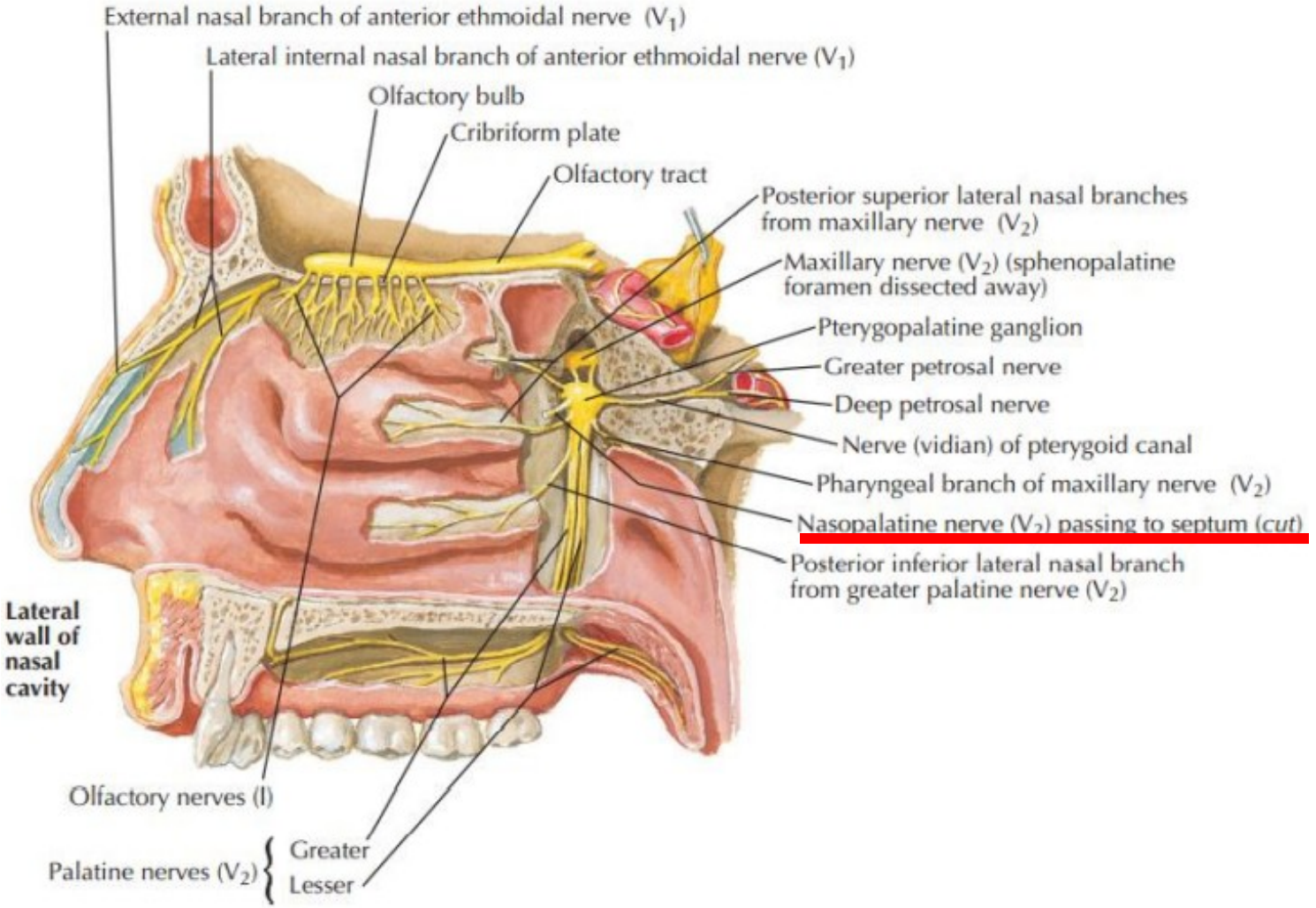


- The **pterygopalatine ganglion** is the largest of the four parasympathetic ganglia in the head.

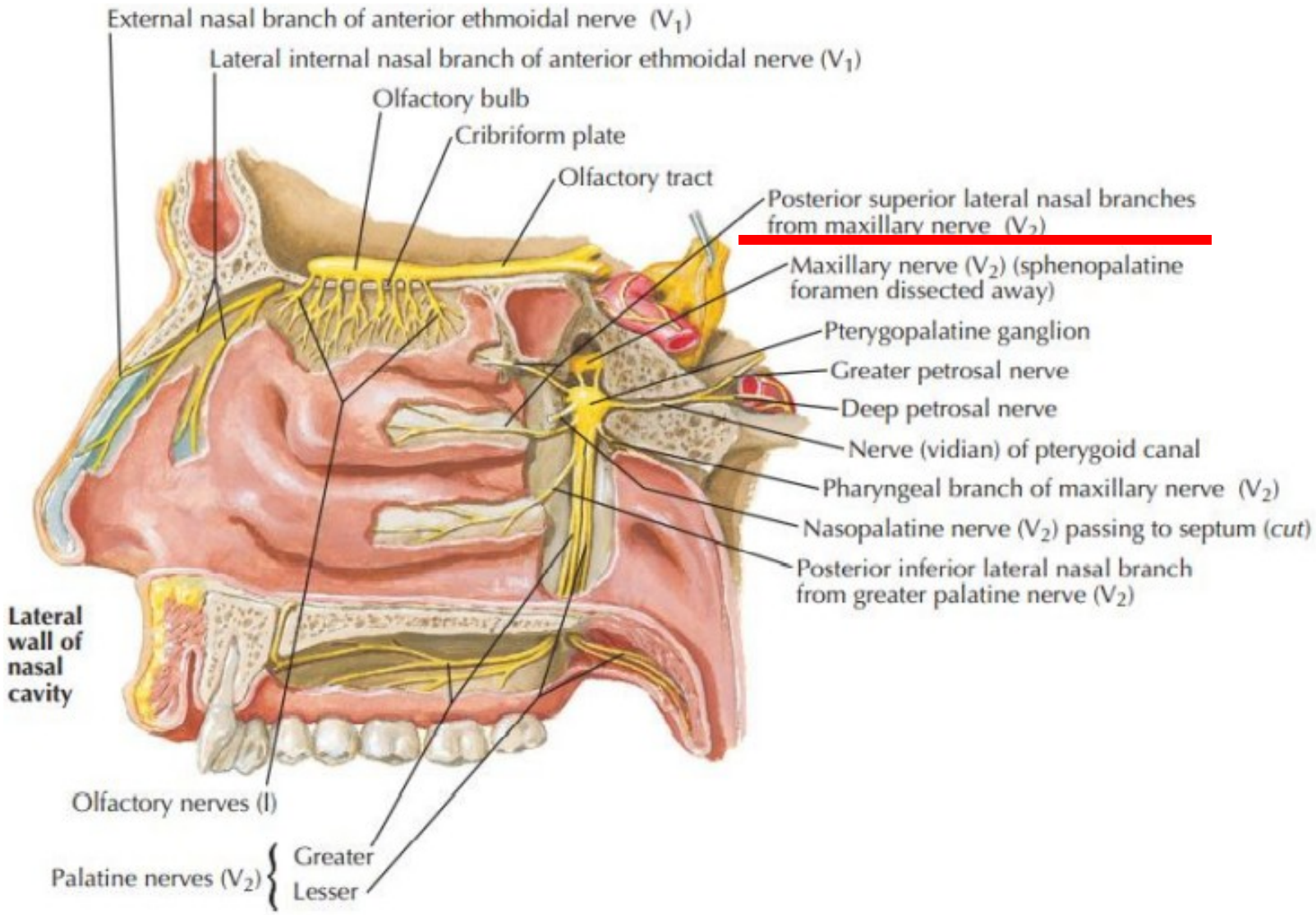
PTERYGOPALATINE GANGLION.

- **Branches:**
- The five branch of pterygopalatine ganglion are distributed to the nose and palate. Every branch carries a mixture of all three kinds of fibers; sensory, parasympathetic and sympathetic.
  1. **Nasopalatine nerve (Sphenopalatine Nerve)**
  2. **Lateral posterior superior nasal nerves (short sphenopalatine nerves)**
  3. **Greater palatine nerves**
  4. **Lesser palatine nerves (two in number)**
  5. **Pharyngeal nerve**

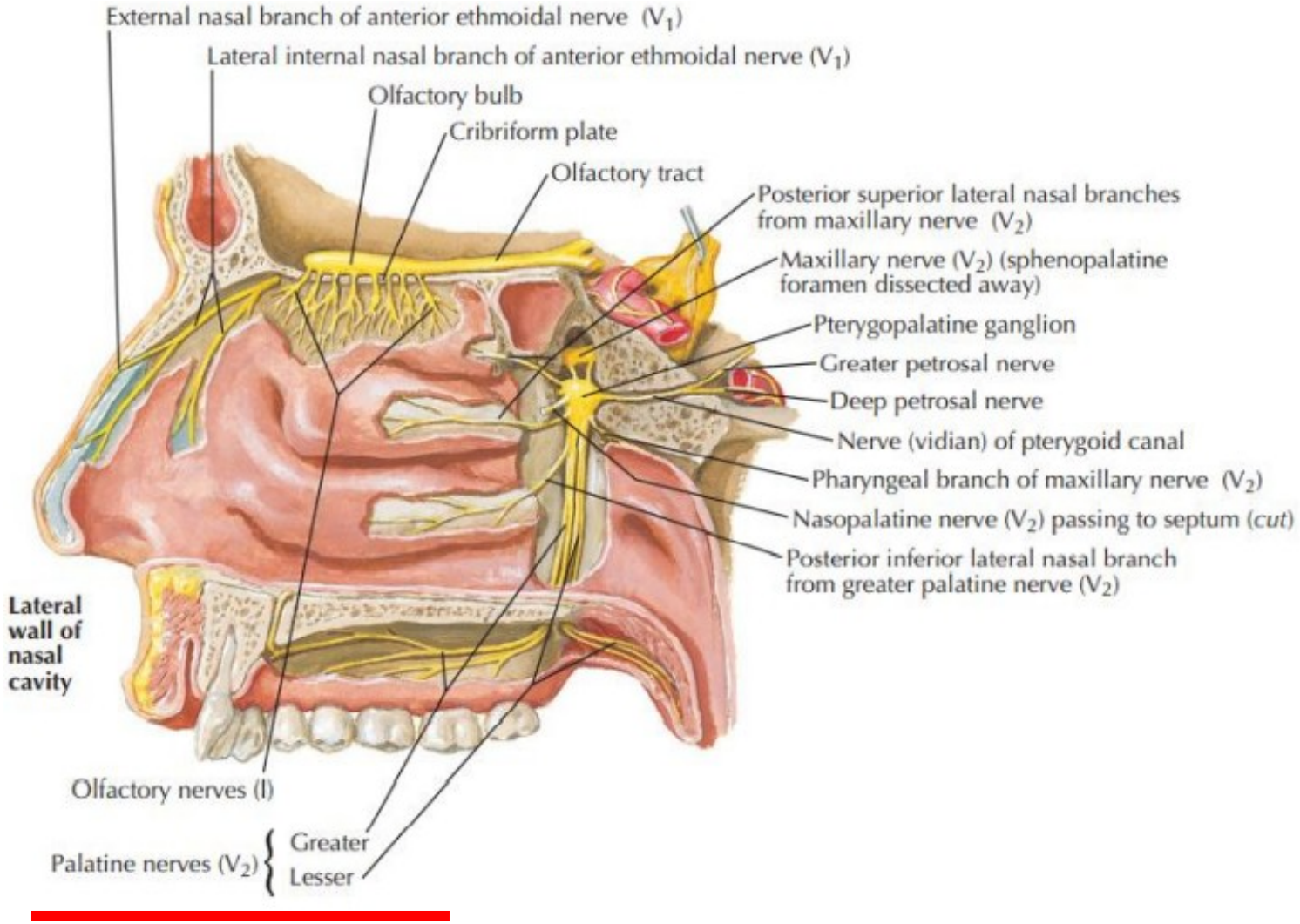




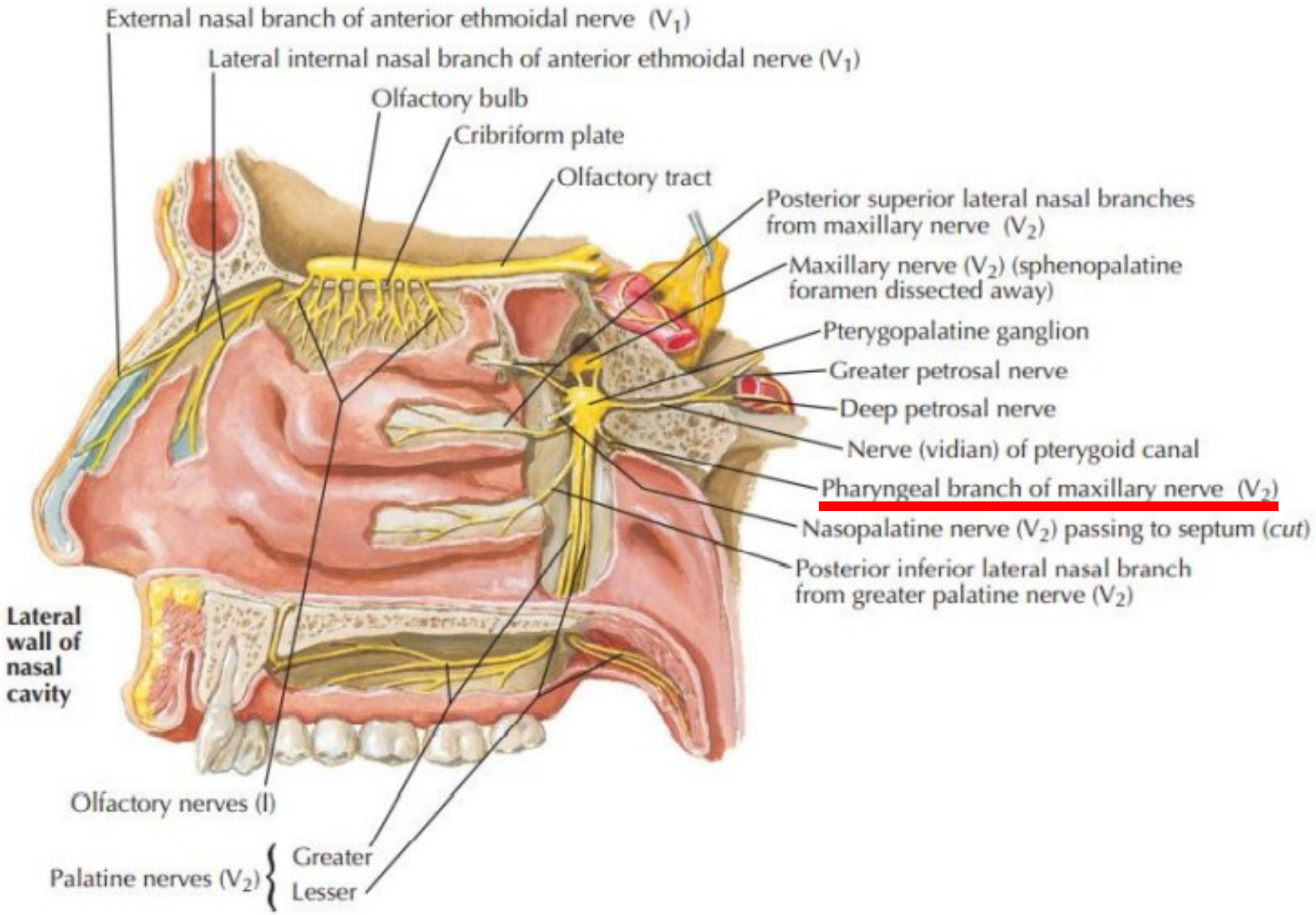
- **Nasopalatine nerve (Sphenopalatine Nerve)----------**
  Sphenopalatine foramen ----is the largest of the nasal nerves and passes anteriorly down the nasal septum, through the incisive canal and fossa in the hard palate to enter the roof of the oral cavity and supply mucosa, gingiva, and glands adjacent to the incisor teeth.



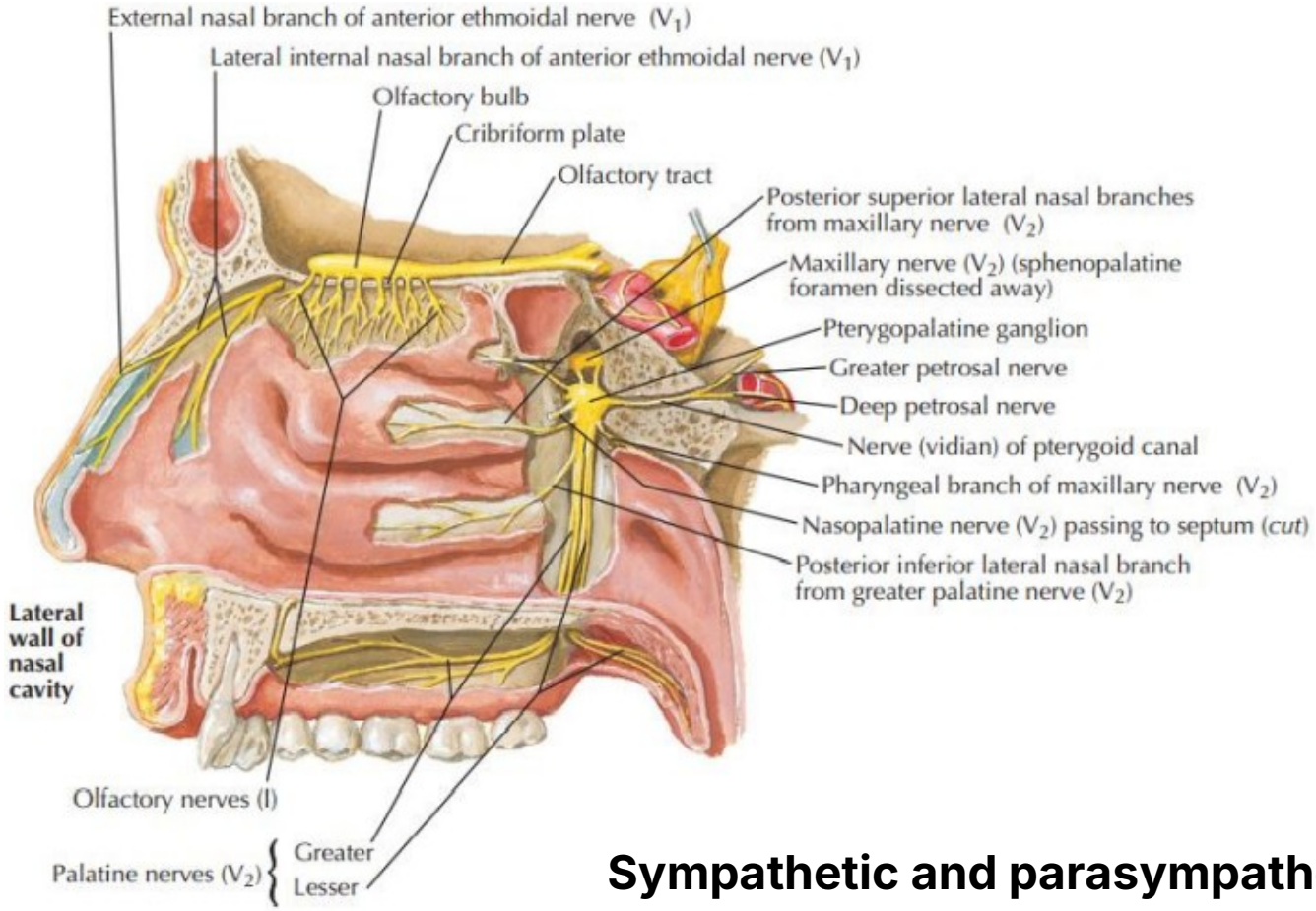
- **Lateral posterior superior nasal nerves (short sphenopalatine nerves)----------** Sphenopalatine foramen----------supplies posterosuperior quadrant of the lateral wall of the nose



- **Greater palatine nerve----------** Greater palatine canal -------greater palatine foramen----------supplies hard palate and posteroinferior quadrant of the -------- lateral wall of nose
- **Lesser palatine nerves (two in number)-------**Passes behind the greater palatine nerve --------lesser palatine foramen-------- supplies soft palate and mucous membrane of palatine tonsil



Pharyngeal nerve----------Palatovaginal canal ----------emerges at the roof of the nose--------supplies the mucous and glands of the upper nasopharynx



Sympathetic and parasympathetic fibres.

Nerve of the pterygoid canal and the pterygopalatine ganglion:

The nerve of the pterygoid canal is formed in the middle cranial fossa by the union of:

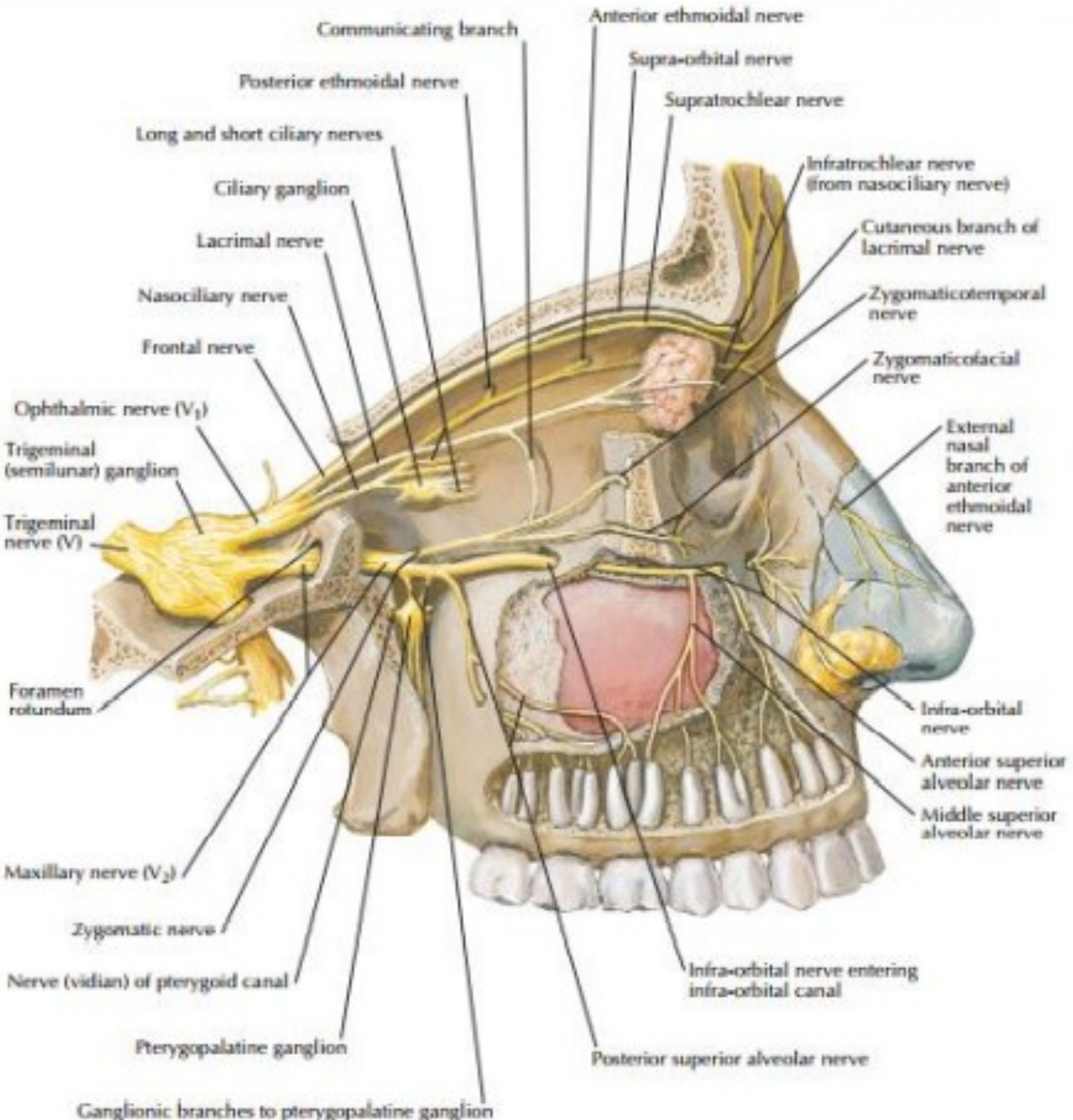
a.the greater petrosal nerve (a branch of the facial nerve [VII]); and

b.the deep petrosal nerve (a branch of the internal carotid plexus).

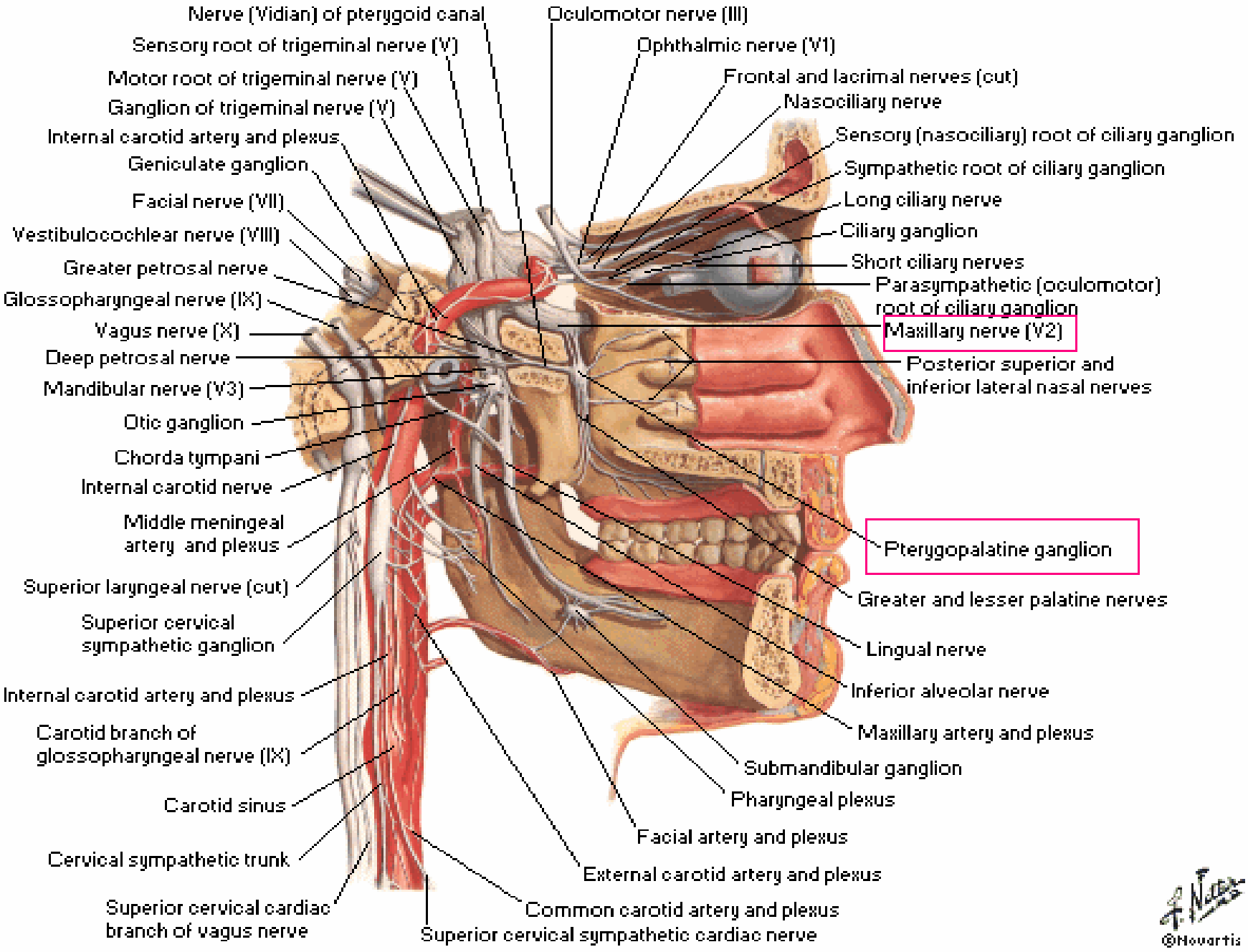
It carries mainly preganglionic parasympathetic and postganglionic sympathetic fibers.

3. MAXILLARY NERVE

- It is a sensory division of the trigeminal nerve.
- Passes through the foramen rotundum.
- Reaches pterygopalatine ganglion.
- Enters inferior orbital fissure.
- Passes through the inferior orbital groove and canal.
- Changes its name to infraorbital nerve as it comes out of the infraorbital foramen

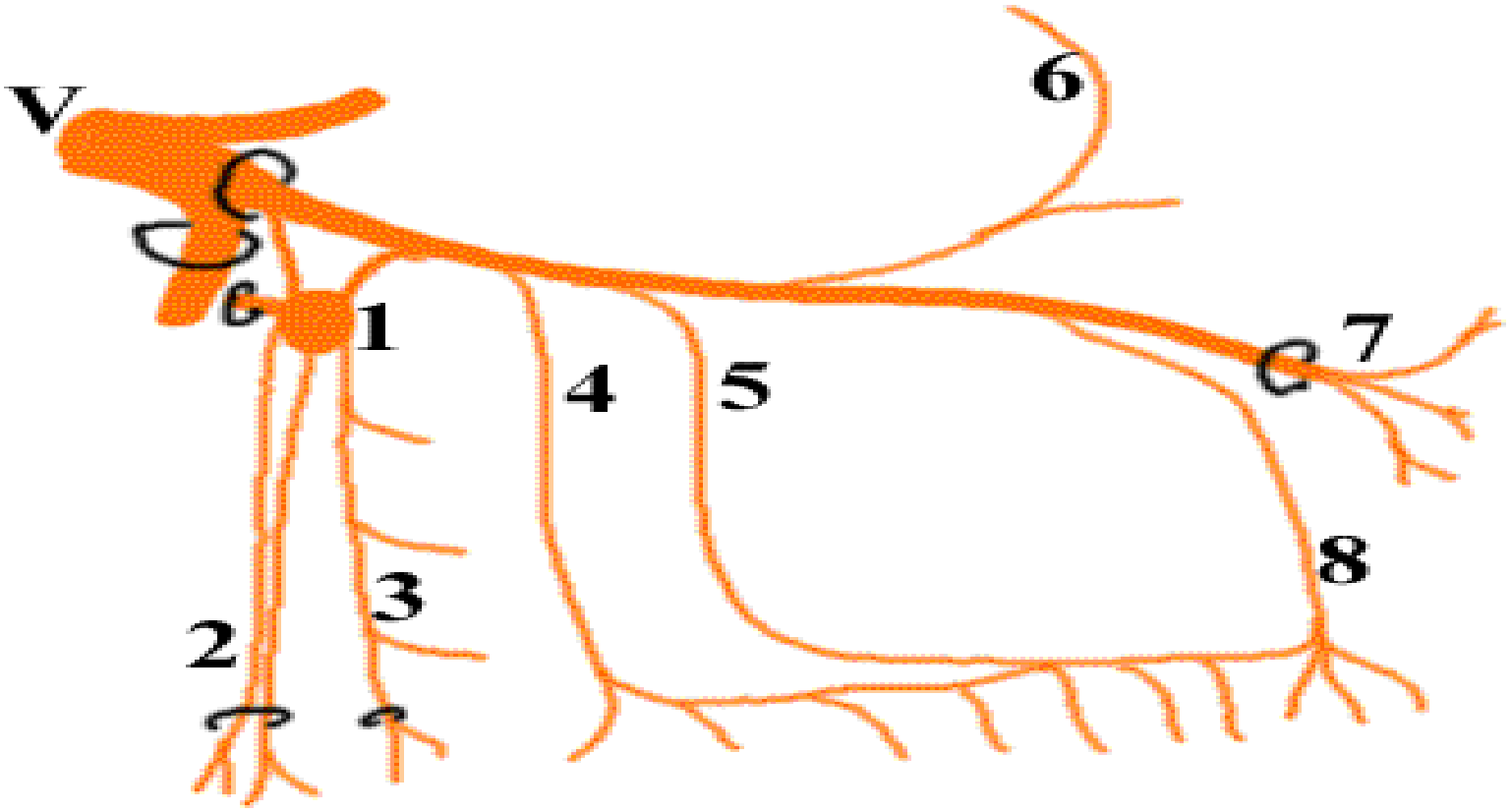


Autonomic Nerves in Head



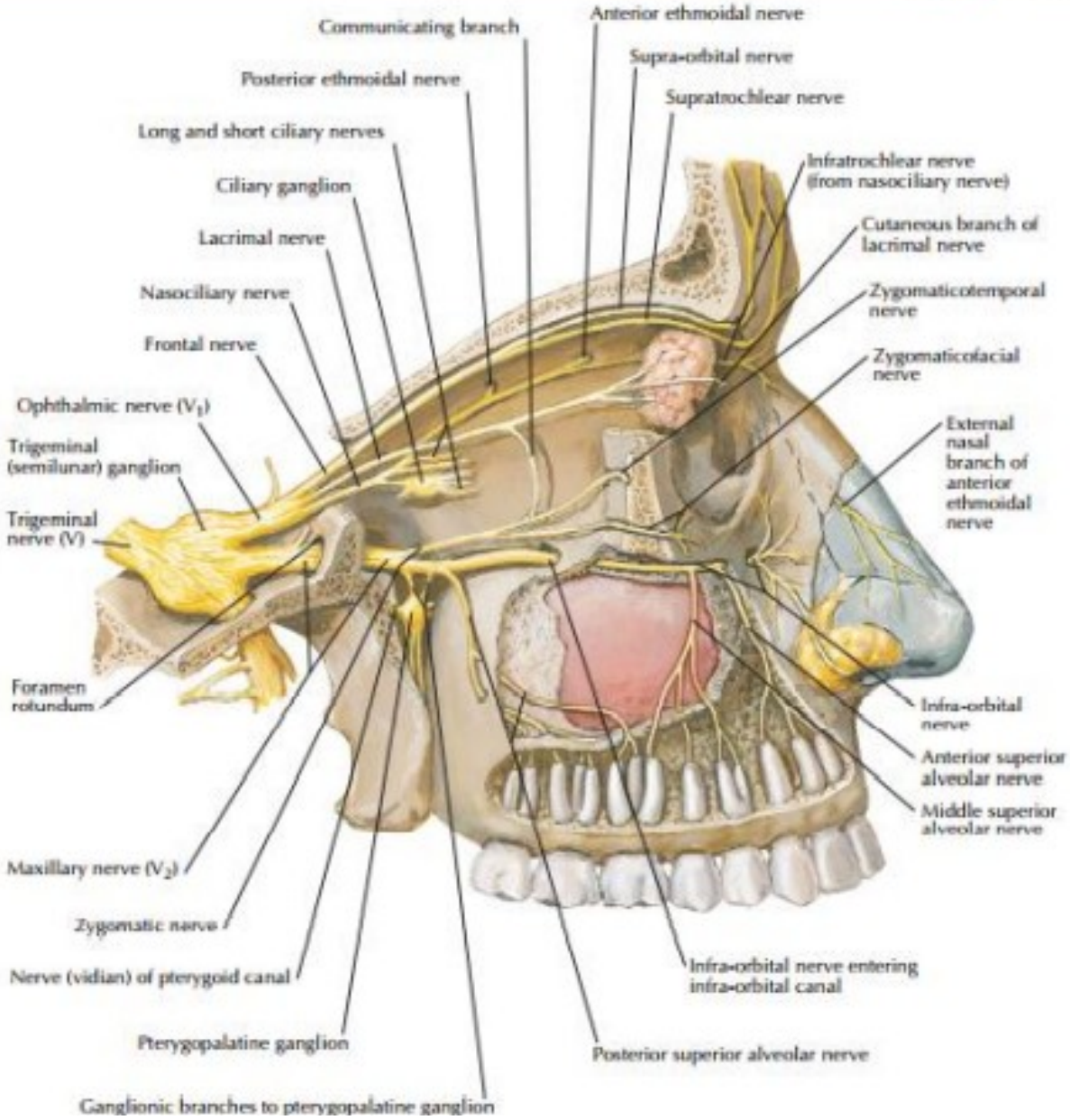
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The maxillary nerve

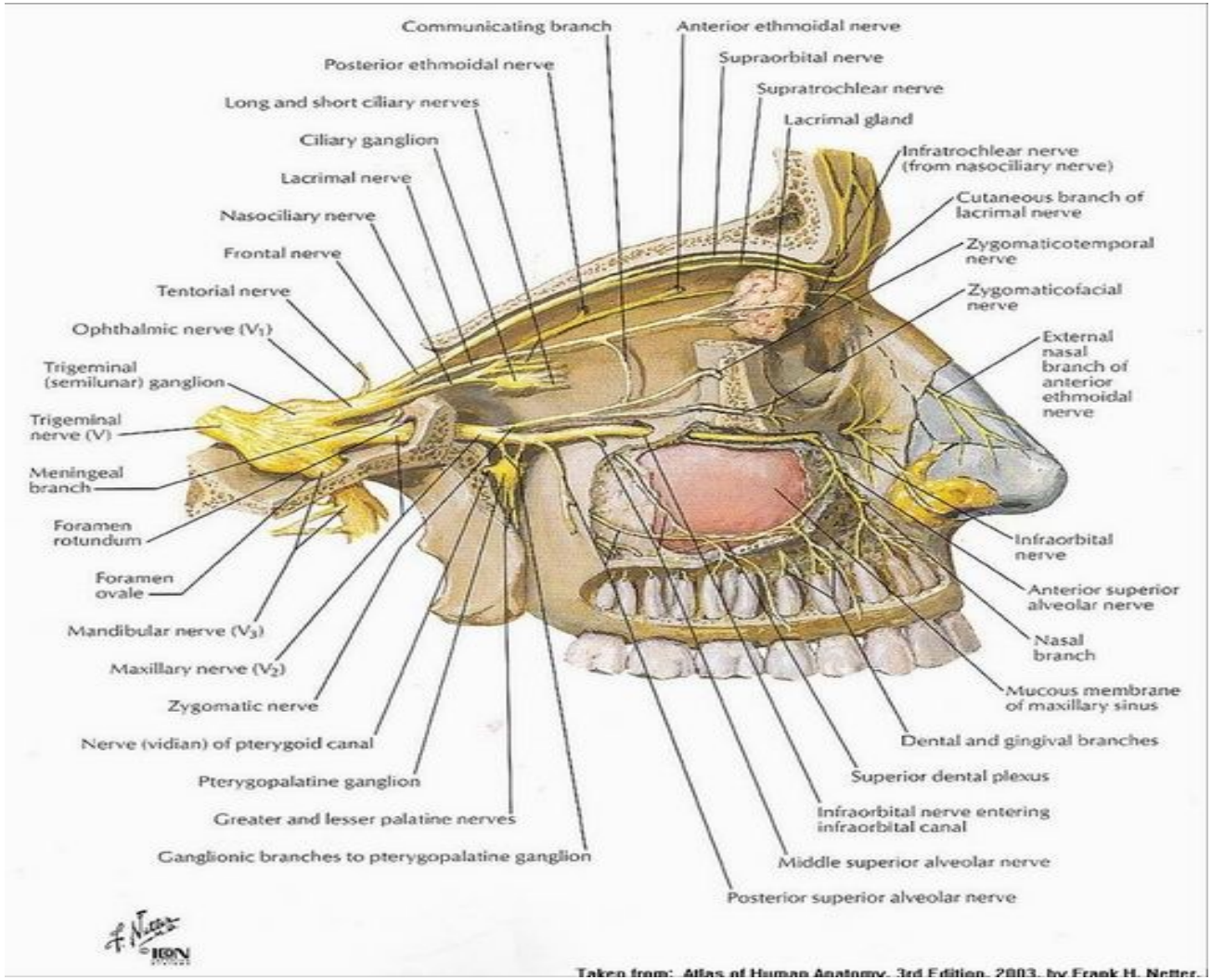


1. pterygopalatine ganglion
2. lesser palatine nerves
3. greater palatine nerve
4. posterior superior alveolar nerve
5. middle superior alveolar nerve
6. zygomaticotemporal branch of zygomatic nerve
7. infraorbital nerve
8. anterior superior alveolar nerve

BRANCHES MAXILLARY NERVE



- In the middle cranial fossa:
  1. Meningeal branch
- In the Pterygopalatine fossa:
  1. **Zygomatic nerves** → inferior orbital fissure→ orbit.
  2. **Posterior superior alveolar nerve** → Pterygo-maxillary fissure.
  3. Two ganglionic branches connecting the maxillary nerve to the Pterygopalatine ganglion.



Taken from: Atlas of Human Anatomy, 3rd Edition, 2003, by Frank H. Netter.

- Postganglionic parasympathetic fibers, arising in the pterygopalatine ganglion, join the general sensory branches of the maxillary nerve [V_2] in the pterygopalatine ganglion, as do postganglionic sympathetic fibers from the carotid plexus, and the three types of fibers leave the ganglion as:

1. Orbital
2. Greater and lesser palatine
3. nasal, and
4. Pharyngeal branches.

OPENINGS IN PTERYGOPALATINE FOSSA.

Walls	Passage	Contents	Passage communicates with
Lateral	Pterygomaxillary fissure	Maxillary artery Maxillary vein Posterior superior alveolar nerve Branch of maxillary nerve	Infratemporal fossa
Medial	Sphenopalatine foramen	Nasopalatine nerve → Pterygopalatine ganglion Lateral posterior superior nasal nerve Sphenopalatine artery → maxillary artery	Nose
Anterior	Inferior orbital fissure	Maxillary nerve Zygomatic nerve → Branch of maxillary nerve	Orbit
Posterior	Palatovaginal canal Foramen Rotundum Pterygoid canal	Pharyngeal branch of maxillary artery Pharyngeal branch of pterygopalatine ganglion Maxillary nerve Nerve to pterygoid canal →Pterygopalatine ganglion Artery to Pterygoid canal → maxillary artery	Pharynx Middle cranial fossa Foramen lacerum
Inferior	Greater palatine foramen Lesser palatine foramen	Greater palatine nerve → Pterygopalatine ganglion Greater palatine artery Maxillary artery Lesser palatine nerve → Pterygopalatine ganglion Lesser palatine arteries → Branches of greater palatine artery	Plate (oral cavity) Palate (oral cavity)

THANK YOU

Fissures

1- inferior orbital fissure
2- pterygomaxillary fissure

Foramina

1- spheno-palatine foramen
2- foramen rotundum

Canals

1- pharyngeal canal
2- Vidian canal
3- pterygopalatine canal

